Formulation of the optimization model in Owl

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1. Introduction

This document describes the mathematical model underlying the optimization algorithms implemented in Owl, which is a Python application optimizing retirement planning using mixed-integer linear programming. Owl is designed for US retirees as it considers US federal tax laws, Medicare premiums, rules for 401k including required minimum distributions, maturation rules for Roth accounts and conversions, social security rules, etc. The goal of these calculations is to optimize the financial aspects of retirement planning, considering the types of savings accounts, federal income tax, contributions, return rates, Medicare premiums, Roth conversions, and desired income amongst many other things.

The approach is described here mathematically and the Python implementation follows the structure and notation presented in this document. The intent of this document is to provide a guide to the source code for any individual desiring to extend the model to other cases. The mathematical description is not tied to any specific mathematical programming language so that it remains as generic as possible and can be solved using different linear solvers.

2. Indices, variables, and parameters

In the next sections, the indices, variables, and parameters are described in detail. Then the model constraints are introduced. For implementation in a linear programming solver, index mapping functions are introduced to map all variables into a single one-dimensional array that is optimized subject to inequality and equality constraints expressed in matrix form. Finally, the constraint matrices are built and so are some useful objective functions.

2.1 Indices

For all indices, we will follow the C array style (starting at 0), rather than the traditional mathematical standard starting at 1. This will facilitate the final sequential mapping of all the variables into a single one-dimensional array, and serve as a direct reference for better understanding the code implementation.

The indices used and their range are defined here, while we also introduce the characteristics and dimensions of the problem. Upper bounds on indices are indicated by the letter N, with the index name as a subscript, e.g., N_i for index i. In other instances, the subscript will indicate that the array depends on subscripts, for example, b_{ikn} is a multidimensional array depending on subscripts i, k, and n.

i Individual. i runs from 0 to $N_i - 1$ where $N_i = 2$ for couples, or $N_i = 1$ for single individuals. The first individual to pass is denoted by i_d while the survivor is i_s.

j Type of savings account. j goes from 0 to $N_j - 1$, for taxable, tax-deferred, and tax-free accounts respectively. Therefore $N_j = 3$.

k Type of asset class. k goes from 0 to $N_k - 1$, for S&P 500, Baa corporate bonds, Treasury notes, and cash, respectively, and therefore $N_k = 4$. More asset classes could be considered at the cost of increasing the complexity of the problem while not generating much more insights.

n Index of the year being modeled. Period being modeled runs from the beginning of year 0 to the first day of the year following year $N_n - 1$, and therefore $N_n + 1$ years are actually considered. Year N_n is the first year following the passing of all individuals in the plan. The time period for all decision variables is annual. For spouses, the end of year $n_d - 1$ is when the first individual is assumed to pass while the survivor will decease at the end of year $N_n - 1$ of the plan.

p Index for long-term capital gains (LTCG) tax rate brackets. $p \in \{0, 1, 2\}$ corresponds to the 0%, 15%, and 20% LTCG rates, respectively, giving $N_p = 3$ brackets.

q Index for income brackets related to Medicare premium adjustments. q goes from 0 to $N_q - 1$, from low to high. At the time of writing, there are 5 Medicare premium steps adjustments in the federal tax code separating $N_q = 6$ income brackets.

t Federal income tax bracket. t goes from 0 to $N_t - 1$, from low to high. At the time of writing, there are $N_t = 7$ federal income tax brackets in the current federal tax code.

2.2 Variables

We will use lowercase roman letters to represent variables. This is done to separate variables from parameters, which will be represented using Greek letters or caligraphic fonts. All variables are assumed to take only non-negative values (≥ 0 inequality). Variables are resolved by the optimizer while parameters are prescribed by the user, historical data, or tax laws.

b_{ijn} Balance for individual i in savings account j at the beginning of year n. When we consider each asset class k, the variable b_{ijkn} is used instead.

d_{in} Deposit of year-n net spending surplus in taxable account of individual i. These deposits are coming from the surplus s_n, distributed to spousal taxable accounts depending on parameter η.

e_n Adjusted standard exemption for year n. This is a variable as the taxable income can sometimes be less that the standard exemption $\bar{\sigma}_n$, leading to a negative taxable income if the inflation-adjusted standard exemption is simply subtracted from the gross taxable income G_n in years of low income.

f_{tn} Amount of taxable ordinary income falling into tax bracket t. The ordinary taxable income is

$$G_n = \sum_t f_{tn}, \tag{2.1}$$

with f_{tn} bound between 0 and the bracket width $\bar{\Delta}_{tn} = \bar{\Gamma}_{tn} - \bar{\Gamma}_{(t-1)n}$. For interpretation, one can introduce the fractional fill f'_{tn} with $f_{tn} := f'_{tn}\bar{\Delta}_{tn}$ and

$$G_n = \sum_t f'_{tn}\bar{\Delta}_{tn}, \tag{2.2}$$

$$0 \leq f'_{tn} \leq 1. \tag{2.3}$$

When considering taxes, the product $\bar{\Delta}_{tn}\theta^x_{tn}$ does not guarantee to be ordered monotonically, and therefore forcing lower brackets to be filled first when being minimized. It is therefore more appropriate to optimize directly in f_{tn}, as the tax rates are progressive. Given that the rates θ^x_{tn} on tax brackets t are increasing monotonically with income, the lower brackets will be filled in first when optimizing. Definitions of $\theta^x_{tn}, \Gamma_{tn}$ and Δ_{tn} are in the section describing the parameters below.

g_n Net spending in year n.

h_{qn} Segment of Medicare bracket q in which the modified adjusted gross income (MAGI) in year $n-2$ is contained. This variable is used to model the income-related monthly adjustment amounts (IRMAA) brackets using special ordered sets of type 1 (SOS1) for the selection on the bracket indicators (as opposed to using a big-$\mathcal{M}$ method). One of the brackets, say q', will contain the MAGI from two years before $\mathbb{G}_{(n-2)}$, and therefore

$$h_{qn} = \delta_(q, q')\mathbb{G}_{(n-2)}, \tag{2.4}$$

where $\delta()$ is a discrete Kronecker function, such that

$$\mathbb{G}_{(n-2)} = \sum_q h_{qn}. \tag{2.5}$$

The role of h will become more clear as the formulation of the MAGI calculation is exposed below.

m_n Medicare costs in year n, including part B premiums and IRMAA income-related adjustments. This does not include part D premiums and adjustments.

s_n Surplus of funds during year n, most likely caused by required minimum distributions (RMDs) or influx of money from big-ticket items (inheritance, gifts, etc.), or from the disposition of fixed assets (sale of residence, restricted stocks, etc.).

w_{ijn} Withdrawal from account j belonging to individual i at the beginning of year n. For the $(j = 1)$ tax-deferred savings account, w_{i1n} is referred to as a distribution for tax purposes as it is a taxable withdrawal, and will always satisfy required minimum distributions.

x_{in} Roth conversion performed by individual i during year n. These events are taxable as ordinary income.

$p_n^{\sigma,\mathrm{lo}}$, $p_n^{\sigma,\mathrm{hi}}$ Excess provisional income above the lower and upper Social Security taxability thresholds, respectively: $p_n^{\sigma,\mathrm{lo}} = \max(0,\, \Pi_n - \mathcal{P}_n^{\mathrm{lo}})$ and $p_n^{\sigma,\mathrm{hi}} = \max(0,\, \Pi_n - \mathcal{P}_n^{\mathrm{hi}})$. These continuous non-negative variables are present only when the solver option `withSSTaxability="optimize"` is selected.

$p_n^{\sigma,\min}$ 50%-zone Social Security taxability allocation, equal to $\min(\Delta\mathcal{P},\, p_n^{\sigma,\mathrm{lo}})$ and bounded in $[0,\, \Delta\mathcal{P}]$. Only present when `withSSTaxability="optimize"` is selected. Together with $p_n^{\sigma,\mathrm{lo}}$ and $p_n^{\sigma,\mathrm{hi}}$, all SS taxability LP variables use the letter p for consistency.

q_{pn} Portion of LTCG and qualified dividends allocated to bracket p in year n, where $p = 0$, 1, 2 correspond to the 0%, 15%, and 20% LTCG rate brackets, respectively. These $N_p = 3$ continuous non-negative LP variables are always present and together partition the total LTCG via $q_{0n}+q_{1n}+q_{2n} \geq Q_n$. The bracket widths are enforced by $q_{0n} \leq \max(0, T_n^{15} - G_n)$ and $q_{0n} + q_{1n} \leq \max(0, T_n^{20} - G_n)$, where T_n^{15} and T_n^{20} are the inflation-adjusted total taxable income thresholds above which the 15% and 20% rates apply, and $G_n = \sum_t f_{tn}$ is the ordinary taxable income (evaluated from the previous SC iteration). Since the cost function is strictly convex ($0 < 0.15 < 0.20$), no binary variables are required: the LP naturally fills the lowest-cost bracket first.

t_n^σ Taxable Social Security amount in year n, bounded in $[0,\ 0.85\sum_i \bar{\zeta}_{in}]$. In the default SC-loop mode this quantity equals $\Psi_n \sum_i \bar{\zeta}_{in}$ and is treated as a fixed parameter; with the solver option `withSSTaxability="optimize"` it becomes an LP decision variable computed exactly from the IRS formula without iteration.

z_*^* Binary variables are all designated by the letter z, but a superscript is used to distinguish the different families. For example, z_{qn}^m represents binary variables associated with Medicare calculations, z_{nz}^x are used for the exclusion constraints, and z_{0n}^σ, z_{1n}^σ encode the two $\min(\cdot,\cdot)$ operations in the SS taxability MIP formulation (see the Social Security taxability paragraph in the Constraints section).

2.3 Parameters

For more easily distinguishing parameters from variables, all parameters are expressed either in Greek letters or uppercase roman letters using caligraphic fonts. Parameter values are either set by the user, historical data, or by the tax code.

β_{ij} Initial balances in savings accounts. These amounts are used to initialize b_{ij0}.

τ_{kn} Annual rate of return for asset class k in year n. A time series of annual return rates for each class of asset. Here, inflation and the rate of return of cash ($k = 3$) are assumed to be the same. In other words, investing in cash yields constant dollars as the return perfectly matches inflation, equivalent to a Treasury inflation-protected security (TIPS) having a real yield of 0%.

γ_n Cumulative inflation at the beginning of year n computed as the product

$$\gamma_n = \prod_{n'=0}^{n-1} (1 + \tau_{3n'}), \tag{2.6}$$

with $\gamma_0 := 1$, and where n' is a dummy index. As the time span of interest goes from the beginning of the first year to the beginning of the year following the last year, variable γ_n will have $N_n + 1$ elements. Parameters indexed for inflation will be indicated by a bar on top as in $\bar{\sigma}_n$. See the next entry for a specific example.

σ_n Standard deduction. It can be adjusted for inflation as follows

$$\bar{\sigma}_n = \sigma_n \gamma_n, \tag{2.7}$$

and can be modified for additional exemptions after 65 of age, for example. It is a simple time series which can include any foreseeable changes in the tax code, or change in filing status due to the passing of one spouse for $n \geq n_d$. The value of $\bar{\sigma}_n$ is an upper bound for variable e_n.

ξ_n Spending profile. This is a time series that multiplies a basis for the desired net spending amount. It is $\xi_n = 1, \forall n$ for a flat profile, or can be a *smile* profile allowing for more money at the start of retirement and modulating it over retirement. Parameter ξ_n can

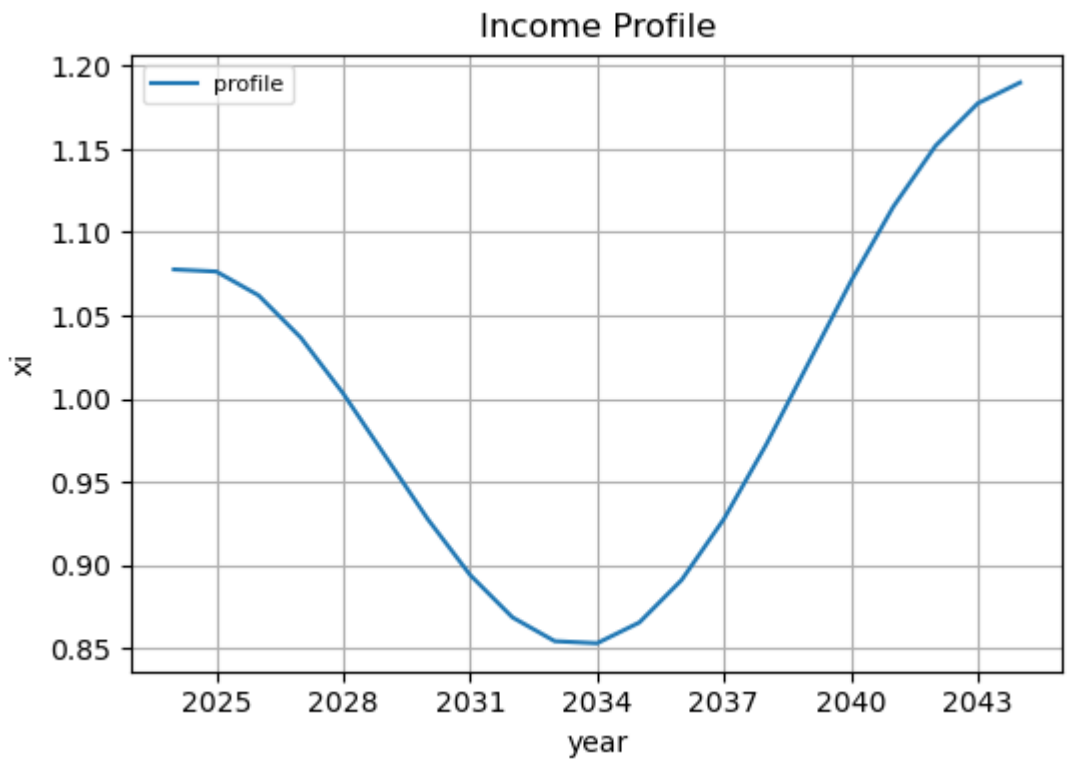

Figure 2.1: Example of a spending profile with 15% cosine factor and a 12% linear profile.

also contain spending adjustments typically made at the passing of one spouse. The *smile* can be implemented using a cosine superimposed over a gentle linear increase such as in

$$\xi_n = 1 + a_1 * \cos(2n\pi/(N_n - 1)) + a_2 n/(N_n - 1), \tag{2.8}$$

and then normalized by factor $N_n/(\sum_n \xi_n)$ to be sum-neutral with respect to a flat profile. Values of $a_1 = 15\%$ and $a_2 = 12\%$ provide curves that are similar to realistic spending profiles reported in the literature. See Fig. 2.1 for an example. At the passing of one spouse, both profiles are reduced by a factor χ for $n \geq n_d$, and the normalizing factor is adjusted accordingly.

χ Factor to reduce spending profile after the passing of one spouse. It is typically assumed to be 0.6. That is, we are assuming that the surviving spouse can live with 60% of the net spending amount that was available to the couple.

ρ_{in} Required minimum distribution for individual i in year n. Expressed in fractions which are determined from IRS tables. These tables are simple if spouses are less than 10 years apart, but a little more complex otherwise, as the age of both spouses need to be taken into account. Current implementation only supports spouses being less that 10 years apart.

An error message is generated if spouses are more than 10 years apart and the calculation is aborted.

Γ_{tn} Bounds for federal income tax brackets. We define $\Gamma_{(-1)n} := 0$, so that Γ_{0n} is the upper bound for the 10% tax bracket in year n. As the filing status can change for couples, and so can the tax code, Γ_{tn} will be changing over n.

Δ_{tn} Difference between upper bound Γ_t and lower bound Γ_{t-1} of a federal income tax bracket,

$$\Delta_{tn} = \Gamma_{tn} - \Gamma_{(t-1)n}. \tag{2.9}$$

Once adjusted for inflation, the taxable income can be expressed as in Eq. (2.1). These data are 7 time series. The filing status changes after the passing of one spouse ($n \geq n_d$) and income tax brackets and differences are adjusted accordingly.

θ^x_{tn} Tax rate for ordinary income tax bracket t in year n. Using N_t time series allows to adjust income tax rates in the foreseeable future. For example, in 2024 the rates (in decimal) are .10, .12, .22, .24, .32, .35, and .37. While these rates were extended indefinitely by Congress in 2025, they could still revert back to 2017 or similar higher rates in the future to .10, .15, .25, .28, .33, .35, and .396. See Eq. (2.25) for its use.

α_{ijkn} Desired asset allocation for savings account j of individual i in assets class k during year n. Allocation ratios come in many flavors as they could be specified globally between individuals and accounts as α_{kn}, for example. When specified by the user, allocation ratios are given two values, one at the beginning of the plan α_{ijk0} and the other at the end $\alpha_{ijkN_{n-1}}$, or α_{ijkn_d} for a spouse passing before the other. Then, intermediate values are interpolated either using a linear relation,

$$\alpha_{ijkn} = a + \frac{n}{N-1}(b-a), \tag{2.10}$$

where N is either n_d or N_n,or using an s-curve as in

$$\alpha_{ijkn} = a + \frac{(b-a)}{2}(\tanh((n-n_1)/n_2)+1), \tag{2.11}$$

where n_1 is the number of years ahead when inflection point will occur, and n_2 is the width (in years) of the transition. Constants n_1 and n_2 can be adjusted by the user. Default values are $n_1 = 15$, and $n_2 = 5$, meaning that the transition center will occur in 15 years, taking place from $15-5$ years to $15+5$ years from now. Using $a = \alpha_{ijk0}$ and $b = \alpha_{ijk(N-1)}$ is an approximation as values of ± 1 are only reached at $\pm\infty$ for a hyperbolic tangent. More precise bounds a' and b' for matching the desired start and end values can be determined by solving a 2×2 system of equations leading to

$$\begin{aligned} a' &= (a - k_{12}b')/k_{11} \\ b' &= (b - (k_{21}/k_{11})a)/(k_{22} - (k_{21}/k_{11})k_{12}), \end{aligned} \tag{2.12}$$

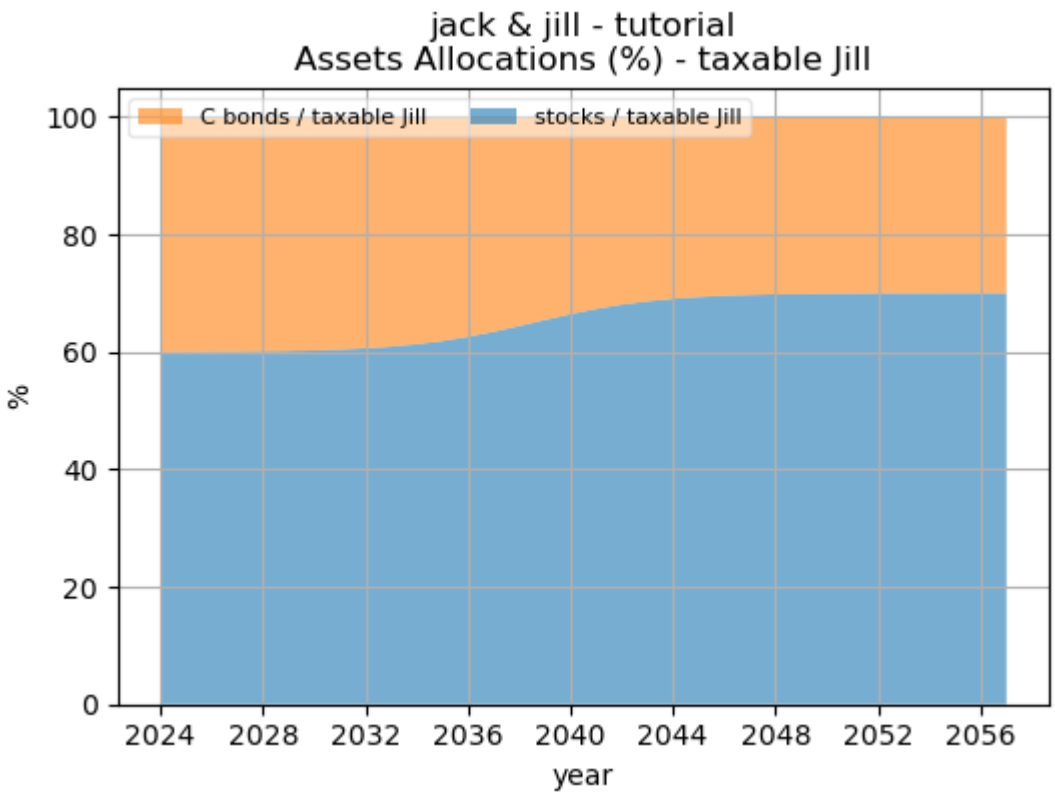

Figure 2.2: Example of an allocation portfolio with 60/40% stocks/bonds transitioning to 70/30% using an s-curve.

where

$$
\begin{aligned}
k_{11} &= \frac{1}{2}(1+\tanh(n_1/n_2)) \\
k_{12} &= \frac{1}{2}(1-\tanh(n_1/n_2)) \\
k_{21} &= \frac{1}{2}(1-\tanh((N-1-n_1)/n_2)) \\
k_{22} &= \frac{1}{2}(1+\tanh((N-1-n_1)/n_2)).
\end{aligned} \tag{2.13}
$$

These interpolation functions allow the allocation ratios to gradually change or *glide* during retirement. Fig. 2.2 provides an example of an *s-curve* gliding allocation ratios.

It is also possible to have a coarser granularity on the portfolio by having an asset allocation scheme defined on a sum of accounts. For example, allocation can be coordinated between accounts leading to α_{ikn}, or even between spouses as α_{kn}. For any of these cases, it is

assumed that weights are always properly scaled so that

$$\begin{aligned} \sum_k \alpha_{ijkn} &= 1, \\ \text{or} \quad \sum_k \alpha_{ikn} &= 1, \\ \text{or} \quad \sum_k \alpha_{kn} &= 1, \end{aligned} \tag{2.14}$$

depending on the scheme selected.

$\mathcal{T}_{ijn}$ When the allocation ratios α_{ijkn} are prescribed, it is sometimes more convenient to express the return rates as

$$\mathcal{T}_{ijn} = \sum_k \alpha_{ijkn} \tau_{kn}. \tag{2.15}$$

$\Lambda^{\pm}_{in}$ Big-ticket item requested by individual i in year n. These are large expenses or influx of money that can be planned. Therefore, $\Lambda^{\pm}$ can be positive (e.g., gift received, inheritance) or negative (e.g., buy a house, large gifts).

λ Allowed deviation from the desired net spending profile during one year. Parameter λ can be better understood as a percentage. If $\lambda = 0.10$, then net spending amount is allowed to vary by up to 10% from the prescribed profile. This parameter is mainly provided for educative purposes.

π_{in} Sum of pension benefits for individual i in year n. These amounts are typically specified along with the ages at which these benefits begin. Pensions can optionally be indexed for inflation and then represented as $\bar{\pi}_{in}$.

ζ_{in} Social security benefits for individual i in year n. Starting age and the passing of one individual for spouses will determine the time series. $\bar{\zeta}_{in}$ is the same series adjusted for inflation.

$\mathcal{P}^{\text{lo}}_n$, $\mathcal{P}^{\text{hi}}_n$ Lower and upper provisional income thresholds for Social Security taxability, following IRS Publication 915. Below $\mathcal{P}^{\text{lo}}_n$ no Social Security is taxable; between $\mathcal{P}^{\text{lo}}_n$ and $\mathcal{P}^{\text{hi}}_n$ up to 50% of benefits enter taxable income; above $\mathcal{P}^{\text{hi}}_n$ up to 85% of benefits are taxable. For single filers $(\mathcal{P}^{\text{lo}}, \mathcal{P}^{\text{hi}}) =$ (\$25,000, \$34,000); for married filing jointly $(\mathcal{P}^{\text{lo}}, \mathcal{P}^{\text{hi}}) =$ (\$32,000, \$44,000). Unlike income-tax brackets, these thresholds are *not* indexed for inflation; the time subscript n reflects only the change in filing status at $n = n_d$.

ϵ_{N_n} Desired amount to leave as a bequest at the end of the final year of the plan, $N_n - 1$, which is the beginning of year N_n. This amount is the after-tax value of the estate for the heirs expressed in today's dollars. See parameter ν for the heirs tax rate.

κ_{ijn} Sum of contributions to savings account j made by individual i during year n. We assume that contributions are made at half-year to better represent periodic contributions made throughout the year. In practice, a contribution amount κ_{ijn} is specified in which case the contribution to each asset class is

$$\kappa_{ijkn} = \alpha_{ijkn} \kappa_{ijn}. \tag{2.16}$$

as savings account balances are assumed to be rebalanced periodically.

ω_{in} Sum of wages earned by individual i during year n. Do not confuse wages ω with withdrawals w.

υ_{in} Other ordinary income for individual i in year n, beyond wages, pension, and Social Security. This is a user-supplied time series (e.g., rental income, alimony) loaded from the household financial plan spreadsheet.

$\mathcal{A}_n^*$ Amounts resulting from the proceeds of liquidation of fixed assets. The liquidation of these fixed assets can generate a taxable income, marked with a superscript x as $\mathcal{A}_n^x$, long-term capital gains, marked with superscript c as $\mathcal{A}_n^c$, or tax-free proceeds marked with a superscript f as $\mathcal{A}_n^f$. These are used to capture the sale of a residence or real estate, cashing restricted stocks, or receiving a lump-sum annuity.

$\mathcal{C}_q$ Cost of Medicare for bracket q of modified adjusted gross income (MAGI) $\mathbb{G}$. This includes Medicare part B premiums and any additional income-related monthly adjusted amount (IRMAA). When adjusted for inflation, this becomes $\bar{\mathcal{C}}_{qn} = \gamma_n \mathcal{C}_q$. There are $N_q = 6$ brackets for IRMAA and therefore 5 step adjustments forming a piecewise constant function.

$\mathcal{D}_n$ Debts payment in year n to be considered in the cash flow. This allows the cash flow to account for a mortgage or car loans for example.

$\mathcal{L}_q$ Upper bounds for brackets used to determine Medicare adjustments based on the modified adjusted gross income (MAGI) $\mathbb{G}$. When adjusted for inflation, this becomes $\bar{\mathcal{L}}_{qn} = \gamma_n \mathcal{L}_q$. While there are 6 brackets for IRMAA adjustments forming a piecewise constant function, $\mathcal{L}_q$ has $N_q - 1$ distinct thresholds as the last element is an arbitrary large number bounding the last bracket. See Fig. 3.1 for a visual representation. Note that $\bar{\mathcal{L}}_{qn}$ needs to be adjusted for inflation and marital status, including adjustments due to the passing of one spouse.

μ Dividend return rate for equities in taxable accounts. Average is about 1.7% for S&P 500.

ν Heirs income tax rate to be applied on the tax-deferred portion of the estate. This is not an estate tax but rather the federal income marginal tax rate that heirs would have to pay on inherited tax-deferred accounts.

ϕ_j Fraction of savings account j that is left to surviving spouse i_s as a beneficiary at the death of individual i_d, the first spouse to pass.

Ψ_n Effective fraction of Social Security benefits subject to income tax in year n, bounded in $[0, 0.85]$. It is computed by the self-consistent loop from the IRS provisional income formula (IRS Publication 915) and enters the LP as a fixed parameter each iteration. See the Social Security taxability paragraph in the Constraints section.

η Spousal ratio for surplus deposits, which goes from 0 to 1, as the fraction that goes to the $i = 1$ spouse's account. Therefore, a surplus s_n in year n will result in a deposit d in the taxable account of individual i as

$$\begin{aligned} d_{0n} &= (1-\eta)s_n \\ d_{1n} &= \eta s_n. \end{aligned} \tag{2.17}$$

This choice is such that we can set a value depending on the surviving individual $\eta = i_s$ for $n \geq n_d$, after the passing of i_d. Default value is $(N_i - 1)/2$, i.e., 0.5 for couples and 0 for single individuals. When the beneficiary of the savings accounts is not the other spouse, i.e., when $\phi_j \neq 1, \forall j$, it is recommended that η be set to i_d so that all surplus get deposited to i_d's accounts, thus avoid loopholes when optimizing for the final bequest.

$\mathcal{M}$ This large constant is used in the so-called big-$\mathcal{M}$ method to implement binary constraints. As this is mainly used around MAGI, a value of about 10^8 should be adequate for most cases. We also use $\mathcal{M}$ to represent the upper bounds of top tax brackets.

2.4 Intermediate variables

We use intermediate variables for conciseness or clarity, but they are ultimately replaced in the final formulation. Intermediate variables are represented in roman uppercase letters, or in double stroke uppercase letters.

G_n Taxable ordinary income in year n. Proceeds from the liquidation of fixed assets taxable as ordinary income, the sum of wages, other ordinary income, pensions, taxable social security benefits, all withdrawals from tax-deferred accounts, including Roth conversions, and gains from securities (i.e., all gains except those from the $k = 0$ equities, which are taxed as capital gains) in the $(j = 0)$ taxable account, including contributions κ, minus the standard deduction e_n,

$$\begin{aligned} G_n &= \mathcal{A}^x_n + \sum_i [\omega_{in} + \upsilon_{in} + \Psi_n \bar{\zeta}_{in} + \bar{\pi}_{in}] - e_n \\ &\quad + \sum_i [w_{i1n} + x_{in}] \\ &\quad + \sum_{i,k\neq 0} [(b_{i0n} - w_{i0n} + d_{in} + 0.5\kappa_{i0n})\alpha_{i0kn}\tau_{kn}] \end{aligned} \tag{2.18}$$

Social security is indexed for inflation. The taxable portion $\Psi_n \sum_i \bar{\zeta}_{in}$ is determined by the self-consistent loop; see the Social Security taxability paragraph in the Constraints section. Pensions can optionally be indexed for inflation. We exclude $k = 0$ to capture only non-equity gains in taxable accounts. These gains are all taxed as ordinary income. Here, we assume that withdrawals and deposits in the taxable account are taking place at the beginning of the year, while contributions, if any, are taking place in mid-year.

Q_n Qualified dividends and long-term capital gains obtained in year n. They only involve dividends occurring in the taxable savings accounts $(j = 0)$ that were obtained from equities $(k = 0)$, or sales of stocks due to withdrawals from taxable savings accounts. For simplicity, we assume that all equity sales only generate long-term capital gains and that all dividends are qualified, resulting in

$$Q_n = \sum_i \alpha_{i00n} \left[(b_{i0n} - w_{i0n} + d_{in} + 0.5\kappa_{i0n})\mu + w_{i0n} \max(0, \tau_{0(n-1)} - \mu) \right]. \tag{2.19}$$

A formulation where only a fraction of dividends are qualified can easily be implemented with the addition of another parameter. Notice that we are using return rates from the

previous year. The first terms on the right-hand side represent dividends generated by equities ($k = 0$) in the ($j = 0$) taxable savings account plus half the yearly contributions. The second term accounts for withdrawals w of equities assumed to have been purchased a year ago. Capital gains are calculated as price appreciation only (total return minus dividend rate) to avoid double taxation of dividends, which are already included in the first term. It does not account for losses, but a market drop would most likely result in stock purchase rather than sale. For withdrawals, we make the assumption of selling the most recent stocks which would not be accurate in situations where the taxable savings account is being depleted slowly. An implementation keeping track of stock purchases and sales is beyond the scope of providing a guide for retirement decisions.

$\mathbb{G}_n$ Modified adjusted gross income or MAGI for year n. We approximate it as

$$\mathbb{G}_n = G_n + Q_n + e_n + (1 - \Psi_n) \sum_i \bar{\zeta}_{in}. \tag{2.20}$$

Because G_n already contains the taxable Social Security term $\Psi_n \sum_i \bar{\zeta}_{in}$, adding back $(1 - \Psi_n) \sum_i \bar{\zeta}_{in}$ restores the full benefit in MAGI, consistent with the IRS definition. This approach ignores additional IRS rules around tax-free interests which are insignificant in most cases.

$\mathbb{I}_n$ Interests and taxable gains earned from taxable account.

$$\mathbb{I}_n = \sum_{i,k\neq 0} [(b_{i0n} - w_{i0n} + d_{in} + 0.5\kappa_{i0n})\alpha_{i0kn}\tau_{kn}]. \tag{2.21}$$

Note that $\mathbb{I}$ can be negative by definition, and τ will need to be clamped to zero if so.

P_n Amount of 10% early withdrawal penalty in year n. The penalty applies only to tax-deferred ($j = 1$) withdrawals before the IRS threshold of age $59\frac{1}{2}$:

$$P_n = 0.10 \sum_i (1 - \mathcal{H}(n - n_{i,59\frac{1}{2}}))w_{i1n}. \tag{2.22}$$

Here, $\mathcal{H}(n - n_{i,59\frac{1}{2}})$ is a Heaviside step function which is 0 or 1, depending on the sign of its argument:

$$\mathcal{H}(x) := \begin{cases} 0 & x < 0 \\ 1 & x \geq 0. \end{cases} \tag{2.23}$$

The parameter $n_{i,59\frac{1}{2}}$ is the year index when individual i turns $59\frac{1}{2}$, or 0 if the individual is already past $59\frac{1}{2}$ at the beginning of the plan. Because the model operates at yearly granularity:

$$n_{i,59\frac{1}{2}} = \max\big(0,\ 59 - Y_0 + y_{\text{obs},i} + \mathbf{1}[m_{\text{obs},i} > 6]\big), \tag{2.24}$$

where Y_0 is the current calendar year and $m_{\text{obs},i} \in \{1, \ldots, 12\}$ is the birth month. The indicator $\mathbf{1}[m > 6]$ adds one year for individuals born in July–December, whose $59\frac{1}{2}$ birthday falls in January–June of the following year. Tax-free ($j = 2$) Roth withdrawals are not subject to this penalty because the 5-year maturation constraints already restrict withdrawals to penalty-free amounts; see the maturation paragraph below.

T_n Amount of income tax paid on taxable ordinary income G_n in year n. This is the taxes paid on ordinary income expressed as the sum of the amounts paid in each tax bracket as

$$T_n = \sum_t f_{tn} \theta^x_{tn}. \tag{2.25}$$

Notice how f_{tn} also defines G_n in Eq. (2.1), and that optimal values of f_{tn} have to minimize T_n regardless of whether the bequest or the desired net spending is being maximized.

U_n Amount of income tax paid on long-term capital gains (LTCG) and qualified dividends in year n. The LTCG tax is computed exactly by the LP bracket variables q_{pn}:

$$U_n = 0.15\, q_{1n} + 0.20\, q_{2n}. \tag{2.26}$$

We assume that qualified dividends and long-term capital gains are taxed at the same preferential rate, which is the case for most situations. For reporting purposes, the effective rate $\psi_n = U_n/Q_n$ is computed after the solve when $Q_n > 0$; it is not a formulation variable.

J_n Net investment income tax (NIIT) paid in year n. The NIIT is 3.8% and applies when modified adjusted gross income $\mathbb{G}_n$ exceeds the statutory threshold $\mathbb{G}^{\text{NIIT}}$ (not indexed for inflation). We compute

$$J_n = 0.038\ \max\left(0,\ \min\left(\mathbb{G}_n - \mathbb{G}^{\text{NIIT}},\ \mathbb{I}_n + Q_n\right)\right), \tag{2.27}$$

where $\mathbb{G}^{\text{NIIT}}$ is \$200k for single filers and \$250k for married filing jointly, with filing status changing after the passing of one spouse ($n \geq n_d$).

3. Formulation with imposed asset allocation ratios

We first present the case where the sums of assets in each savings accounts b_{ijn} are known and for which we assume a prescribed asset allocation ratios. The amount in each asset class k for b_{ijkn} is simply obtained from $\alpha_{ijkn} b_{ijn}$ in this case. This formulation assumes that the accounts are always balanced. This is a reasonable assumption given the auto-balancing feature offered by many financial service providers and robot advisers.

The benefit of this approach is that it has less variables and that only the sums of all asset classes in each savings account need to be considered. The rate of return of the account is then simply the product of the account balance with the sum of the rates of return weighted according to the desired allocation ratio. This approach allows us to eliminate k by summing over it and rewrite equations such as

$$\sum_k b_{ijk(n+1)} = \sum_k b_{ijkn}(1+\tau_{kn}) + \ldots, \tag{3.1}$$

for the annual evolution of account balances from year n to year $n+1$ as the simpler expression

$$\begin{aligned} b_{ij(n+1)} &= b_{ijn} \sum_k \alpha_{ijkn}(1+\tau_{kn}) + \ldots, \\ &= b_{ijn}(1+\mathcal{T}_{ijn}) + \ldots, \end{aligned} \tag{3.2}$$

where

$$\mathcal{T}_{ijn} := \sum_k \alpha_{ijkn}\tau_{kn}. \tag{3.3}$$

This is a consequence that the allocation ratios are normalized to unity, i.e.,

$$\sum_k \alpha_{ijkn} = 1. \tag{3.4}$$

In this formulation where the α_{ijkn} are prescribed, we will use $\mathcal{T}_{ijn}$ to add the market returns to the savings balances.

3.1 Constraints

Required minimum distributions (RMDs) Withdrawals from the ($j = 1$) tax-deferred savings accounts must be larger or equal than the required minimum distributions, and therefore,

$$w_{i1n} - \rho_{in} b_{i1n} \geq 0. \tag{3.5}$$

As b_{ijn} are the balances at the beginning of year n, they are also the balances at December 31 of the previous year, which is the amount from which the IRS bases the RMDs. Eq. (3.5) has to hold for each year n and each individual i, and therefore, there are $i \times N_n$ such equations (although trivial when $\rho_{in} = 0$). These constraints avoid paying up to 25% penalty on amounts not withdrawn when RMDs are required. Note that aggregate rules need to be considered separately as this approach only considers the sum of assets in a class with similar tax treatment (e.g., IRA and 401k).

Income tax brackets Taxable ordinary income is divided in tax brackets as defined in Eq. (2.1), and therefore

$$0 \leq f_{tn} \leq \bar{\Delta}_{tn}, \tag{3.6}$$

where each income tax bracket width Δ_{tn} has been adjusted for inflation.

Standard exemption The standard exemption is constrained by

$$0 \leq e_n \leq \bar{\sigma}_n. \tag{3.7}$$

Variable e_n is required for accounting for years when the taxable ordinary income is smaller that the standard exemption.

Withdrawal limits We introduce another set of constraints that might look unnecessary, but can help convergence, and prevent overdrafts during the year of passing of spouse i_d. As withdrawals and conversions are at the beginning of the year we impose that

$$w_{ijn} + \delta(j,1) x_{in} \leq b_{ijn}, \tag{3.8}$$

which just states that account balances need to be larger than withdrawals and possible Roth conversions.

Posthumous account activities For cases with spouses, no withdrawal, Roth conversion, or deposit should occur in the accounts of the passed individual i_d:

$$\begin{aligned} w_{i_d jn} &= 0, \\ d_{i_d n} &= 0, \\ x_{i_d n} &= 0, \\ & \quad \forall j \in 0, \ldots, N_j - 1 \\ & \quad \forall n \in n_d, \ldots, N_n - 1. \end{aligned} \tag{3.9}$$

Roth conversions Roth conversions cannot be larger than the balance at the beginning of the year in the account:

$$x_{ijn} \leq b_{i1n}. \tag{3.10}$$

This constraint, however, is naturally satisfied when $b_{ijn} \geq 0$ non-negativity bounds are enforced. Additional maximum Roth conversion constraints x_{max} can be imposed by the user. For a single individual, the previous equation becomes

$$x_{in} \leq \min(b_{i1n}, x_{max}). \tag{3.11}$$

For couples, the cap applies to each individual's conversions:

$$x_{in} \leq x_{max}, \quad i \in \{0, 1\}. \tag{3.12}$$

As these equations involve a variable and a parameter in the min function, they are coded as separate constraints.

Initial balances The initial balances β_{ij} are one of the main inputs of the model. The initial savings account balances are imposed through the constraints

$$b_{ij0} = \beta_{ij}. \tag{3.13}$$

At this point, we assume that all accounts are balanced according to the desired allocation ratios α_{ijk0}.

Roth 5-year maturation and conversion ladder Two IRS rules govern penalty-free access to Roth funds. (1) Roth *contributions* can always be withdrawn tax- and penalty-free. (2) Roth *conversions* can be withdrawn penalty-free only after a 5-year holding period, with each conversion carrying its own clock; *earnings* additionally require age $59\frac{1}{2}$. Because of rule (2), any Roth withdrawal allowed by the maturation constraint is already penalty-free, so no 10% early-withdrawal penalty is assessed on w_{i2n}.

The maturation constraint (row group I_8 in the matrix appendix) enforces this rule:

$$b_{i2n} - w_{i2n} \;\geq\; \sum_{k=1}^{5} \Gamma_k \, x_{i,n-k} + \sum_{k=1}^{5} (\Gamma_k - 1) \, \kappa_{i2,n-k}, \quad \Gamma_k = \prod_{\ell=1}^{k} \max(1,\, 1 + \mathcal{T}_{i2,n-\ell}), \tag{3.14}$$

forcing immature conversions (those made within the last 5 years) and recent contribution gains to remain in the account. Historical amounts ($n - k < 0$) use a conservative 10%/yr assumption.

This constraint enables the *Roth conversion ladder*, a strategy popular in the FIRE (Financial Independence, Retire Early) community. An early retiree converts tax-deferred funds in years $n, n+1, \ldots$ and, 5 years later, withdraws the corresponding Roth amounts penalty-free. The optimizer discovers this strategy automatically once the Roth penalty is removed and the AMO constraint is relaxed for pre-$59\frac{1}{2}$ years (see the AMO discussion below).

Cash-flow surplus When both spouses are alive, surplus s_n gets deposited in the taxable accounts according to variable η as described in Eq. (2.17),

$$d_{in} = [\delta(i,0)(1-\eta) + \delta(i,1)\eta] s_n. \tag{3.15}$$

Otherwise, for $n \geq n_d$, variable η gets redefined as $\eta = \delta(1, i_s)$. Surplus can be caused by large influx of money coming from big-ticket items, compulsory RMDs, or the disposition of fixed assets.

Account balances Contributions are assumed to be made at half-year to better represent periodic contributions made throughout the year. As we already mentioned, the account balance at the end of a year is the same as the balance at the beginning of the following year. Changes include contributions κ, distributions and withdrawals w, conversions x, surplus deposits d, and growth τ on the account through the year. For each spouse i, we track each savings account j separately, and tax-deferred accounts are coupled with the corresponding tax-free account through Roth conversions.

The timing of Roth conversions, withdrawals, and deposits brings additional coupling between these variables, and is worth a detailed discussion. First, the financial aspects, and then the algorithmic ones. For the former, some financial advisors would recommend making Roth conversions at the beginning of the year, while making withdrawals at the end. Obviously, financial simulators would always yield higher numbers when using this scenario, as the moneys needed to pay the regular bills stayed in the bank until the end of the year. More realistically, however, it would be more accurate to assume withdrawals at mid-year, to better represent evenly distributed withdrawals. So, financially, conversions at the beginning of the year, and withdrawals at mid-year make good sense. Conversions are also typically best when timed with market downturns, which are obviously not always at the beginning of the year.

Now, let's look at the optimization side of these transactions. During years of positive returns, a direct withdrawal from the tax-deferred account at mid-year will always be unfavorable when compared to a Roth conversion at the beginning of the year, followed by a tax-free withdrawal later in the same year. This is because the second scenario involves gains which are tax-free over the half-year, while the first one does not. Moving account withdrawals at the beginning of the year, and the conversions in mid-year can only solve part of this artificial bias under specific conditions.

Moving all transactions at the beginning of the year can also cause undesired transactions. To solve these spurious scenarios, it would be desirable to make the following at-most-one (AMO) exclusions between surplus s_n, withdrawals w_{ijn}, and conversions x_{in}:

$$\left(\sum_i w_{i0n} + \sum_i w_{i2n}\right) \quad \texttt{AMO} \quad s_n,$$
$$\sum_i x_{in} \quad \texttt{AMO} \quad \sum_i w_{i2n},$$

for $j \neq 1$, i.e., for all withdrawals except those from tax-deferred accounts. The AMO relation can also be represented by a logical NAND operator. The first exclusion prevents surplus deposits when withdrawals from taxable or tax-free accounts occur from either spouse, while the second exclusion prevents simultaneous Roth conversions and tax-free withdrawals from either spouse. To favor tax-deferred withdrawals in most reasonable situations, we implement these exclusions by introducing binary variables $z^x_{n\ell} \in \{0,1\}$ with $\ell \in \{0,1,2,3\}$ for each year n (shared across both spouses). The first pair of binary variables (z^x_{n0}, z^x_{n1}) enforces the surplus-withdrawal exclusion:

$$\begin{aligned}
\epsilon z^x_{n0} &\leq \sum_i (w_{i0n} + w_{i2n}) \leq \mathcal{M} z^x_{n0}, \\
\epsilon z^x_{n1} &\leq s_n \leq \mathcal{M} z^x_{n1}, \\
0 &\leq z^x_{n0} + z^x_{n1} \leq 1,
\end{aligned} \tag{3.16}$$

where ϵ is a small positive number (e.g., 0.001) to ensure that when a binary variable is active, the corresponding continuous variable is non-zero. However, the AMO exclusion does not require

the lower bound on the variables as both values can be zero. The second pair of binary variables (z^x_{n2}, z^x_{n3}) enforces the Roth conversion-tax-free withdrawal exclusion:

$$\begin{aligned} \epsilon z^x_{n2} \leq & \sum_i x_{in} \leq \mathcal{M} z^x_{n2}, \\ \epsilon z^x_{n3} \leq & \sum_i w_{i2n} \leq \mathcal{M} z^x_{n3}, \\ 0 \leq & z^x_{n2} + z^x_{n3} \leq 1. \end{aligned} \tag{3.17}$$

Here, $\mathcal{M}$ is a large number such as 10^9, just slightly larger than what x, w, and s can possibly be. The lower bound can be ignored here as well for the same reason. Note that the Roth conversion/tax-free withdrawal AMO (Eq. 3.17) is *relaxed* for years $n < \max_i(n_{i,59\frac{1}{2}})$: before any individual reaches age $59\frac{1}{2}$, it is legitimate to simultaneously withdraw mature 5-year-old conversions and initiate new ladder rungs, so enforcing the AMO in those years would block the Roth conversion ladder. The AMO is reinstated once all individuals have passed $59\frac{1}{2}$, preventing artificial round-trips.

Another approach could be to perform Roth conversions at mid-year, while withdrawals could be made at the beginning of the year, and surplus deposits, if needed due to RMDs or receiving large sums of money, could be made at the end of the year. Let's formulate this approach in more detail and investigate for potential problems. Timing controls which terms get multiplied by the rate of return $(1 + \mathcal{T}_{ijn})$. Therefore, our current choice would yield

$$\begin{aligned} b_{ij(n+1)} \quad = \quad & [b_{ijn} - w_{ijn} + 0.5\kappa_{ijn}](1 + \mathcal{T}_{ijn}) + [\delta(j,2) - \delta(j,1)]x_{in}(1 + \mathcal{T}_{ijn}/2) \\ & + \delta(j,0)d_{in} + 0.5\kappa_{ijn}, \end{aligned} \tag{3.18}$$

where we use discrete Kronecker δ functions for selecting the specific accounts involved in Roth conversions. These conversions are made such that asset allocation ratios in the sending and receiving accounts are unchanged.

Bringing all variables to the left-hand side, this gets rewritten as

$$\begin{aligned} & b_{ij(n+1)} - (b_{ijn} - w_{ijn})(1 + \mathcal{T}_{ijn}) \\ - & [\delta(j,2) - \delta(j,1)]x_{in}(1 + \mathcal{T}_{ijn}/2) - \delta(j,0)d_{in} \quad = \quad \kappa_{ijn}(1 + \mathcal{T}_{ijn}/2). \end{aligned} \tag{3.19}$$

When $j = 0$, this equation introduces a path to shelter negative returns by performing an over-withdrawal from the taxable account at the beginning of the year followed by a deposit in the same account at the end of the year. This can be removed by using another binary variable, thus making these events exclusive by using the same strategy as Eqs. (3.16) and (3.17).

A much simpler approach, while not so natural, is to move all transactions to be synchronous at the beginning or at the end of the year thus avoiding undesirable movements of funds. If we select the beginning of the year, except for contributions, this leads to

$$b_{ij(n+1)} - [b_{ijn} + \delta(j,0)d_{in} - w_{ijn} + (\delta(j,2) - \delta(j,1))x_{in}](1 + \mathcal{T}_{ijn}) \quad = \quad \kappa_{ijn}(1 + \mathcal{T}_{ijn}/2).$$

This is the current approach used in Owl, coupled with binary variables excluding simultaneous overwithdrawals and deposits, and simultaneous Roth conversions and withdrawals from tax-free accounts.

Net spending For calculating the net spending g_n, we consider the cash flow of all withdrawals, wages, other ordinary income, social security and pension benefits, proceeds from liquidation of fixed assets, and big-ticket items. Then we subtract potential surplus s_n and all taxes, penalties, debts, and Medicare premiums paid:

$$\begin{aligned} g_n &= \sum_i [\omega_{in} + \upsilon_{in} + \bar{\zeta}_{in} + \bar{\pi}_{in}] + \sum_{i,j} w_{ijn} + \sum_i \Lambda^{\pm}_{in} + \sum_{*=x,c,f} \mathcal{A}^*_n \\ &\quad -s_n - P_n - T_n - U_n - J_n - m_n - \mathcal{D}_n. \end{aligned} \tag{3.20}$$

Notice how big-ticket items $\Lambda^{\pm}$ contribute directly to the cash flow. Replacing intermediate variables and bringing all variables to the left-hand side, we get

$$\begin{aligned} g_n - \sum_{i,j} w_{ijn} + 0.1 \sum_i (1 - \mathcal{H}(n - n_{i,59\frac{1}{2}})) w_{i1n} & \\ +m_n + s_n + \sum_t f_{tn}\theta^x_{tn} + U_n &= \sum_{*=x,c,f} \mathcal{A}^*_n + \sum_i \Lambda^{\pm}_{in} \\ &\quad + \sum_i [\omega_{in} + \upsilon_{in} + \bar{\zeta}_{in} + \bar{\pi}_{in}] \\ &\quad -\mathcal{D}_n - J_n. \end{aligned} \tag{3.21}$$

Notice that we do not consider market losses or tax-loss harvesting, and we do not track individual stock purchases over the years. We also treat J_n as a parameter as it is computed through a self-consistent loop.

LTCG bracket constraints The LTCG tax $U_n = 0.15\, q_{1n} + 0.20\, q_{2n}$ is determined by three LP bracket allocation variables q_{pn} ($p \in \{0, 1, 2\}$). The partition constraint

$$q_{0n} + q_{1n} + q_{2n} \geq Q_n \tag{3.22}$$

ensures the allocations cover the total LTCG Q_n. The bracket upper-bound constraints prevent overflow into higher-rate brackets. Because LTCG is stacked on top of ordinary taxable income $G_n = \sum_t f_{tn}$, the room available in each bracket depends on how much of the threshold is already occupied by ordinary income. Using the value of G_n from the previous SC iteration as a parameter (consistent with the treatment of Ψ_n, $\mathcal{J}_n$, and $\mathcal{M}_n$), the constraints are

$$q_{0n} \leq \max(0,\ T^{15}_n - G_n), \tag{3.23}$$

$$q_{0n} + q_{1n} \leq \max(0,\ T^{20}_n - G_n), \tag{3.24}$$

where T^{15}_n and T^{20}_n are the inflation-adjusted total taxable income thresholds above which the 15% and 20% LTCG rates apply. These constraints converge together with the SC loop. Since the cost is convex (0% < 15% < 20%), no binary variables are needed: the LP naturally allocates gains to the cheapest bracket first.

We want the net spending to be predictable and smooth. For that purpose, we use

$$g_n/\bar{\xi}_n = g_0/\bar{\xi}_0, \tag{3.25}$$

where the net spending is adjusted for inflation and where we use the time series of parameter ξ_n, allowing for additional adjustments to the overall desired spending. Note that $\bar{\xi}_0 = \xi_0$ as $\gamma_0 = 1$.

This profile is used to lower the desired net spending amount by a reduction factor χ after the passing of one spouse and/or to allow for more realistic spending profiles, such as the *smile* profile described above. Eq. (3.25) can be rewritten as

$$g_n \xi_0 - g_0 \bar{\xi}_n = 0, \tag{3.26}$$

for the constraints to be enforced. Once g_0 is determined, the whole time series of net spending is determined. When using slack variable λ, the spending profile is then implemented as inequality constraints

$$\begin{aligned} g_n \bar{\xi}_0 - g_0(1-\lambda)\bar{\xi}_n &\geq 0, \\ -g_n \bar{\xi}_0 + g_0(1+\lambda)\bar{\xi}_n &\geq 0. \end{aligned} \tag{3.27}$$

In this case, the objective function will need to consider the sum over all years when optimizing net spending.

Taxable ordinary income We connect the two definitions for G_n stated above in Eqs. (2.1) and (2.18),

$$\begin{aligned} \sum_t f_{tn} &= \mathcal{A}_n^x + \sum_i [\omega_{in} + \upsilon_{in} + \Psi_n \bar{\zeta}_{in} + \bar{\pi}_{in}] \\ &\quad + \sum_i [w_{i1n} + x_{in}] \\ &\quad + \sum_{i,k\neq 0} [(b_{i0n} - w_{i0n} + d_{in} + 0.5\kappa_{i0n})\alpha_{i0kn}\tau_{kn}] - e_n, \end{aligned} \tag{3.28}$$

and re-arrange to move variables to the LHS as follows

$$\begin{aligned} e_n + \sum_t f_{tn} - \sum_i [w_{i1n} + x_{in}] & \\ - \sum_{i,k\neq 0} [(b_{i0n} - w_{i0n} + d_{in})\alpha_{i0kn}\tau_{kn}] &= \mathcal{A}_n^x + \sum_i [\omega_{in} + \upsilon_{in} + \Psi_n \bar{\zeta}_{in} + \bar{\pi}_{in}] \\ &\quad + 0.5 \sum_{i,k\neq 0} \alpha_{i0kn}\tau_{kn}\kappa_{i0n}. \end{aligned} \tag{3.29}$$

Social Security taxability Under IRS Publication 915, the fraction of Social Security benefits subject to income tax is determined by the *provisional income* (PI), defined as

$$\Pi_n = \mathbb{G}_n - \tfrac{1}{2} \sum_i \bar{\zeta}_{in}. \tag{3.30}$$

The IRS piecewise formula gives the taxable SS amount as

$$\mathcal{T}_n = \begin{cases} 0 & \Pi_n < \mathcal{P}^{\text{lo}}, \\ 0.5\,(\Pi_n - \mathcal{P}^{\text{lo}}) & \mathcal{P}^{\text{lo}} \leq \Pi_n < \mathcal{P}^{\text{hi}}, \\ \min\Big(0.85 \sum_i \bar{\zeta}_{in},\ 0.5\,(\mathcal{P}^{\text{hi}} - \mathcal{P}^{\text{lo}}) + 0.85\,(\Pi_n - \mathcal{P}^{\text{hi}})\Big) & \Pi_n \geq \mathcal{P}^{\text{hi}}, \end{cases} \tag{3.31}$$

so that $\Psi_n = \mathcal{T}_n \,/\, \sum_i \bar{\zeta}_{in} \in [0,\, 0.85]$.

Introducing the excess provisional income above each threshold,

$$p_n^{\sigma,\mathrm{lo}} = \max(0,\, \Pi_n - \mathcal{P}_n^{\mathrm{lo}}), \qquad p_n^{\sigma,\mathrm{hi}} = \max(0,\, \Pi_n - \mathcal{P}_n^{\mathrm{hi}}), \tag{3.32}$$

the three cases of Eq. (3.31) collapse into the single expression

$$\mathcal{T}_n = \min\Big(0.85 \sum_i \bar{\zeta}_{in},\; 0.50 \min\Big(\sum_i \bar{\zeta}_{in},\, \min(\Delta\mathcal{P},\, p_n^{\sigma,\mathrm{lo}})\Big) + 0.85\, p_n^{\sigma,\mathrm{hi}}\Big), \tag{3.33}$$

where $\Delta\mathcal{P} = \mathcal{P}^{\mathrm{hi}} - \mathcal{P}^{\mathrm{lo}}$ is the bracket width (\$12,000 for MFJ; \$9,000 for single filers). The inner $\min(\sum_i \bar{\zeta}_{in},\, \cdot\,)$ guards against the unusual case $\sum_i \bar{\zeta}_{in} < \Delta\mathcal{P}$, in which SS benefits are too small to fill the 50%-bracket entirely. Because the clipped excess $\min(\Delta\mathcal{P},\, p_n^{\sigma,\mathrm{lo}}) \le \Delta\mathcal{P}$, whenever $\sum_i \bar{\zeta}_{in} \ge \Delta\mathcal{P}$ the inner min is automatically satisfied and Eq. (3.33) simplifies to

$$\mathcal{T}_n = \min\Big(0.85 \sum_i \bar{\zeta}_{in},\; 0.5\, p_n^{\sigma,\min} + 0.85\, p_n^{\sigma,\mathrm{hi}}\Big), \qquad p_n^{\sigma,\min} = \min(\Delta\mathcal{P},\, p_n^{\sigma,\mathrm{lo}}). \tag{3.34}$$

This condition holds for virtually all retirees, since combined annual SS benefits nearly always exceed \$12,000 (MFJ) or \$9,000 (single).

Because Ψ_n enters the LP only as a constant multiplier of the fixed series $\bar{\zeta}_{in}$ on the right-hand side of Eq. (3.29), it is treated as a *parameter* rather than a decision variable. It is updated between LP solves in the self-consistent (SC) loop: after each solve, $\mathbb{G}_n$ is computed from the current solution, Π_n is evaluated, and Ψ_n is refreshed via Eq. (3.31). A 30% damping blend $\Psi_n \leftarrow 0.3\, \Psi_n^{\mathrm{new}} + 0.7\, \Psi_n$ prevents oscillation near the bracket boundaries. To avoid unnecessary LP rebuilds, Ψ_n is only updated when $\max_n |\Delta\Psi_n| > 10^{-3}$. The SC loop terminates when the change in the objective function (net spending or bequest) falls below a prescribed tolerance.

Alternatively, when the solver option `withSSTaxability="optimize"` is set, all max/min operations in Eqs. (3.32)–(3.34) are encoded exactly within the LP as a mixed-integer program (MIP) using the continuous variables $p_n^{\sigma,\mathrm{lo}}$, $p_n^{\sigma,\mathrm{hi}}$, $p_n^{\sigma,\min}$, and t_n^{σ}, together with two binary variables z_{0n}^{σ} and z_{1n}^{σ} per year n for which $\sum_i \bar{\zeta}_{in} > 0$; in years with no Social Security income all four continuous variables and both binaries are trivially fixed to zero and excluded from the MIP enumeration. The non-negativity of $p_n^{\sigma,\mathrm{lo}}$ and $p_n^{\sigma,\mathrm{hi}}$ combined with the lower-bound constraints $p_n^{\sigma,\mathrm{lo}} \ge \Pi_n - \mathcal{P}_n^{\mathrm{lo}}$ and $p_n^{\sigma,\mathrm{hi}} \ge \Pi_n - \mathcal{P}_n^{\mathrm{hi}}$ enforce the $\max(0, \cdot)$ operations; because maximizing spending minimizes taxes, the optimizer pushes these variables to their lower bounds at the optimum. The two $\min(\cdot, \cdot)$ operations in Eq. (3.34) are formulated via big-$\mathcal{M}$ lower-bound constraints. For $p_n^{\sigma,\min} = \min(\Delta\mathcal{P},\, p_n^{\sigma,\mathrm{lo}})$, the upper bounds

$$p_n^{\sigma,\min} \le \Delta\mathcal{P}, \qquad p_n^{\sigma,\min} \le p_n^{\sigma,\mathrm{lo}} \tag{3.35}$$

are complemented by binary-controlled lower bounds that force the optimizer to the tighter constraint:

$$p_n^{\sigma,\min} - \mathcal{M}\, z_{0n}^{\sigma} \;\ge\; \Delta\mathcal{P} - \mathcal{M}, \qquad p_n^{\sigma,\min} - p_n^{\sigma,\mathrm{lo}} + \mathcal{M}\, z_{0n}^{\sigma} \;\ge\; 0. \tag{3.36}$$

Analogously, for $t_n^{\sigma} = \min\big(0.85 \sum_i \bar{\zeta}_{in},\; 0.5\, p_n^{\sigma,\min} + 0.85\, p_n^{\sigma,\mathrm{hi}}\big)$:

$$t_n^{\sigma} \le 0.85 \sum_i \bar{\zeta}_{in}, \qquad t_n^{\sigma} \le 0.5\, p_n^{\sigma,\min} + 0.85\, p_n^{\sigma,\mathrm{hi}}, \tag{3.37}$$

$$t_n^{\sigma} - \mathcal{M}\, z_{1n}^{\sigma} \;\ge\; 0.85 \sum_i \bar{\zeta}_{in} - \mathcal{M}, \qquad t_n^{\sigma} - 0.5\, p_n^{\sigma,\min} - 0.85\, p_n^{\sigma,\mathrm{hi}} + \mathcal{M}\, z_{1n}^{\sigma} \;\ge\; 0. \tag{3.38}$$

In this formulation t_n^σ is an LP variable rather than the fixed parameter $\Psi_n \sum_i \bar{\zeta}_{in}$ used in SC-loop mode. Moving t_n^σ to the left-hand side, Eq. (3.29) becomes

$$\begin{aligned} e_n + \sum_t f_{tn} - t_n^\sigma - \sum_i [w_{i1n} + x_{in}] & \\ - \sum_{i,k\neq 0} [(b_{i0n} - w_{i0n} + d_{in})\alpha_{i0kn}\tau_{kn}] &= \mathcal{A}_n^x + \sum_i [\omega_{in} + \upsilon_{in} + \bar{\pi}_{in}] \\ &\quad +0.5 \sum_{i,k\neq 0} \alpha_{i0kn}\tau_{kn}\kappa_{i0n}, \end{aligned} \tag{3.39}$$

and $\Psi_n = t_n^\sigma / \sum_i \bar{\zeta}_{in}$ is recovered from the LP solution for reporting. There is no circular dependency: provisional income Π_n is constructed from non-SS ordinary income, Roth conversions, Q_n dividends, and the fixed half of SS ($0.5 \sum_i \bar{\zeta}_{in}$), so t_n^σ does not appear in Π_n.

Medicare brackets and costs Annual Medicare costs m_n include the income-related monthly adjusted amount commonly known as IRMAA. As this additional adjustment is a piecewise constant function, it can be computed using binary variables and mixed-integer linear programming. In the current tax code, this adjustment depends on the modified adjusted gross income (MAGI) $\mathbb{G}$ from 2 years earlier. For the MAGI, we use $\mathbb{G}_{(n-2)} = G_{(n-2)} + Q_{(n-2)} + e_{(n-2)} + (1 - \Psi_{(n-2)}) \sum_i \bar{\zeta}_{i(n-2)}$, i.e., gross taxable ordinary income plus dividends, plus standard exemption and the non-taxable portion of social security from 2 years ago. If the plan has its oldest individual currently either 64 or 65 years old, MAGI values for previous years must be provided by the user to complete the calculations.

Including regular premiums, there are $N_q = 6$ brackets of MAGI indexed for inflation defined with annual cumulative Medicare costs of $\bar{\mathcal{C}}_{qn} = \gamma_n \mathcal{C}_q$ also adjusted for inflation. These segments are separated by $N_q - 1$ thresholds $\bar{\mathcal{L}}_{qn} = \gamma_n \mathcal{L}_q$. Points $(\bar{\mathcal{L}}_{qn}, \bar{\mathcal{C}}_{qn})$ form a piecewise constant function that could be modeled using a special ordered set of type 1 (SOS1). Points $(\mathcal{L}_q, \mathcal{C}_q)$ are shown in Fig. 3.1. When $\mathbb{G}_{(n-2)} \leq \bar{\mathcal{L}}_{0n}$, the value $\bar{\mathcal{C}}_{0n}$ represents the (inflation-adjusted) base premium for Medicare part B in year n, while higher q values include the IRMAA additional costs.

To model the IRMAA brackets we introduce binary variables $z_{qn}^m \in \{0, 1\}$ (one per bracket) and continuous variables $h_{qn} \geq 0$ for the MAGI assigned to the selected bracket q. Big-$\mathcal{M}$ is used only for the upper bound of the top bracket (see below). We impose an SOS1 selection on binary z_{qn}^m as

$$\sum_q z_{qn}^m = 1, \tag{3.40}$$

and the disaggregation constraints

$$\sum_{q=0}^{N_q-1} h_{qn} = \mathbb{G}_{(n-2)}, \tag{3.41}$$

$$\begin{aligned} 0 \leq h_{0n} &\leq \bar{\mathcal{L}}_{0n} z_{0n}^m, \\ \bar{\mathcal{L}}_{(q-1)n} z_{qn}^m \leq h_{qn} &\leq \bar{\mathcal{L}}_{qn} z_{qn}^m, \quad q \in \{1, \ldots, N_q - 2\}, \\ \bar{\mathcal{L}}_{(N_q-2)n} z_{(N_q-1)n}^m \leq h_{(N_q-1)n} &\leq \mathcal{M}\gamma_n z_{(N_q-1)n}^m. \end{aligned} \tag{3.42}$$

for all $n \in \{n_m, \ldots, N_n - 1\}$, where n_m is the index year when Medicare starts for any individual i in the plan. The upper bound on the last bracket is necessary: when $z_{(N_q-1)n}^m = 0$ (the top bracket

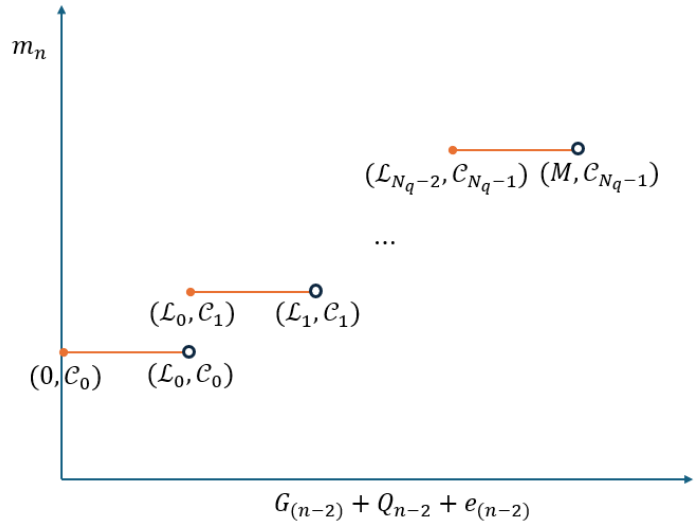

Figure 3.1: Modeling a piecewise constant monotonically increasing function. The x-axis represents the MAGI from two years ago, while m_n is the Medicare adjusted premiums. A binary variable z_q^m is associated with each segment q.

is not selected), the lower bound alone only gives $h_{(N_q-1)n} \geq 0$, so the solver could assign positive MAGI to an inactive bracket. The constraint $h_{(N_q-1)n} \leq \mathcal{M}\gamma_n z^m_{(N_q-1)n}$ forces $h_{(N_q-1)n} \leq 0$ when $z^m_{(N_q-1)n} = 0$; with $h_{qn} \geq 0$ we obtain $h_{(N_q-1)n} = 0$ for the inactive bracket. When $z^m_{(N_q-1)n} = 1$, $\mathcal{M}\gamma_n$ is chosen large enough to not bind (e.g. of order 10^8 in real units). As this applies to each individual eligible for Medicare, values $\bar{\mathcal{L}}_{qn}$ and $\bar{\mathcal{C}}_{qn}$ can include the accounting details for when both spouses are eligible and/or alive. For each year $n \geq n_m$, we can then express the Medicare costs m_n as

$$m_n = \sum_{q=0}^{N_q-1} z^m_{qn} \bar{\mathcal{C}}_{qn}, \tag{3.43}$$

as $z^m_{qn} = \delta(q, q')$ can only select a unique bracket q'. For $n \geq 2$, Eq. (3.41) can be expanded using

Eqs. (2.18) and (2.19) as

$$\begin{aligned}
\sum_{q=0}^{N_q-1} h_{qn} \quad = \quad & \mathcal{A}^x_{(n-2)} + \mathcal{A}^c_{(n-2)} + \sum_i [\omega_{i(n-2)} + \upsilon_{i(n-2)} + \bar{\zeta}_{i(n-2)} + \bar{\pi}_{i(n-2)}] \\
& -e_{(n-2)} - \sum_i [w_{i1(n-2)} + x_{i(n-2)}] \\
& - \sum_i \left[(b_{i0(n-2)} - w_{i0(n-2)} + d_{i(n-2)}) \left(\sum_{k\neq 0} [\alpha_{i0k(n-2)} \tau_{k(n-2)}] + \mu \alpha_{i00(n-2)} \right) \right] \\
& + \sum_i [w_{i0(n-2)} \alpha_{i00(n-2)} \max(0, \tau_{0\,\max(0,n-3)} - \mu)] \\
& +0.5 \sum_i \left[\kappa_{i0(n-2)} \left(\sum_{k\neq 0} [\alpha_{i0k(n-2)} \tau_{k(n-2)}] + \mu \alpha_{i00(n-2)} \right) \right].
\end{aligned}$$

Collecting variables on the left-hand side yields

$$\begin{aligned}
& \sum_{q=0}^{N_q-1} h_{qn} + e_{(n-2)} + \sum_i [w_{i1(n-2)} + x_{i(n-2)}] \\
& \qquad + \sum_i \Big[(b_{i0(n-2)} - w_{i0(n-2)} + d_{i(n-2)}) \\
& \qquad\qquad \cdot \left(\sum_{k\neq 0} [\alpha_{i0k(n-2)} \tau_{k(n-2)}] + \mu \alpha_{i00(n-2)} \right) \Big] \\
& \qquad - \sum_i [w_{i0(n-2)} \alpha_{i00(n-2)} \max(0, \tau_{0\,\max(0,n-3)} - \mu)] \\
& \qquad = \mathcal{A}^x_{(n-2)} + \mathcal{A}^c_{(n-2)} + \sum_i [\omega_{i(n-2)} + \upsilon_{i(n-2)} + \bar{\zeta}_{i(n-2)} + \bar{\pi}_{i(n-2)}] \\
& \qquad\qquad + 0.5 \sum_i \Big[\kappa_{i0(n-2)} \left(\sum_{k\neq 0} [\alpha_{i0k(n-2)} \tau_{k(n-2)}] + \mu \alpha_{i00(n-2)} \right) \Big].
\end{aligned} \tag{3.44}$$

Eq. (3.41) is expanded using Eqs. (2.18) and (2.19) to yield a linear equality in the decision variables; it is implemented directly in the constraint matrix. For indices at the beginning of the time sequence, we use $\tau_{0,\max(0,n-x)}$, when $x > n$. Note that $G_{(n-2)}$ already includes the taxable portion of social security (via $\Psi_n \bar{\zeta}_{in}$), and the additional $(1 - \Psi_{(n-2)})\bar{\zeta}_{i(n-2)}$ term restores the non-taxable portion so that MAGI uses total social security benefits.

For plans where the year index when Medicare first starts $n_m < 2$, we will request and use user-provided values for the first-years MAGIs as $\mathbb{G}_{(-1)}$ and possibly $\mathbb{G}_{(-2)}$ and set

$$\sum_{q=0}^{N_q-1} h_{qn} = \mathbb{G}_{(n-2)}, \qquad n \in \{n_m, \ldots, 1\}, \tag{3.45}$$

where $n_m \geq 0$. Using Eq. (3.42), we can also check for bounds and add constraints directly on z_{q0}^m and possibly z_{q1}^m.

4. Mapping of decision variables

At this point, one can use one of the many algebraic modeling languages such as AMPL, GAMS, MOSEK, AIMMS, and Gurobi, and code the equations above using that language, but most of these applications are proprietary and require a license and additional software installation. These languages allow the problem to be stated at a high level and steps to cast the problem in a form suitable for solution are performed automatically. There are also object-oriented language extensions, such as Python's Pyomo and PuLP that can ease the process of solving these problems. For completeness, however, we present here a simple index mapping approach that allows solving this problem using a generic linear programming solver.

Using a simple interface for mapping sparse objects to dense ones, the approach described here has been successfully tested with both the HiGHS open-source solver and the MOSEK proprietary solver. To cast the problem in a form suitable for a linear programming solver, we will use a single block vector represented by the array $y[q()]$ with index-mapping functions $q()$. While this process can be achieved using slicing and reshaping in some programming languages, we will present a generic approach suitable for most programming languages. The detailed approach presented here also allows us to determine the size of the problem to solve. We proceed alphabetically for all variables, and continue to use the convention of having index 0 for representing the first element.

To bring all variables in a single block vector, we will simply use two generic index mapping functions defined as

$$q_*(C, \ell_1, \ell_2, \ell_3; N_1, N_2, N_3) := C + \ell_1 N_2 N_3 + \ell_2 N_3 + \ell_3, \tag{4.1}$$

and

$$q_C(C, N_1, N_2, N_3) := C + N_1 N_2 N_3, \tag{4.2}$$

with the constraint that $0 \leq \ell_i < N_i$.

Account balances (b) For storing the savings account balances appropriately, variable b_{ijn} needs to have one more entry $(N_n + 1)$ to store the end-of-life estate value. Therefore, we use

$$y[q_b(i, j, n)] = b_{ijn}, \tag{4.3}$$

where

$$q_b(i, j, n) = q_*(C_b, i, j, n; N_i, N_j, N_n + 1), \tag{4.4}$$

and where n exceptionally runs from 0 to N_n inclusively, and therefore q_b runs from $C_b = 0$ to $C_d - 1$, where

$$C_d = q_C(C_b, N_i, N_j, N_n + 1) = N_i N_j (N_n + 1).$$

Surplus deposits (d) For surplus deposits in the taxable savings accounts d_{in} we will use

$$y[q_d(i,n)] = d_{in}, \tag{4.5}$$

where

$$q_d(i,n) = q_*(C_d, i, n, 0; N_i, N_n, 1), \tag{4.6}$$

with q_d running from C_d to $C_e - 1$, where

$$C_e = q_C(C_d, N_i, N_n, 1) = N_i(N_j(N_n + 1) + N_n).$$

Standard exemption (e) For the standard exemption e_n we will use

$$y[q_e(n)] = e_n, \tag{4.7}$$

where

$$q_e(n) = q_*(C_e, n, 0, 0; N_n, 1, 1) = C_e + n, \tag{4.8}$$

with q_e running from C_e to $C_f - 1$, where

$$C_f = q_C(C_e, N_n, 1, 1) = N_i(N_j(N_n + 1) + N_n) + N_n.$$

Tax bracket amounts (f) For tax bracket amounts f_{tn} we will use

$$y[q_f(t,n)] = f_{tn}, \tag{4.9}$$

where

$$q_f(t,n) = q_*(C_f, t, n, 0; N_t, N_n, 1) \tag{4.10}$$

with q_f running from C_f to $C_g - 1$, where

$$C_g = q_C(C_f, N_t, N_n, 1) = N_i(N_j(N_n + 1) + N_n) + (N_t + 1)N_n.$$

Net spending (g) For net spending g_n we will use

$$y[q_g(n)] = g_n, \tag{4.11}$$

where

$$q_g(n) = q_*(C_g, n, 0, 0; N_n, 1, 1) = C_g + n, \tag{4.12}$$

with q_g running from C_g to $C_h - 1$, where

$$C_h = q_C(C_g, N_n, 1, 1) = N_i(N_j(N_n + 1) + N_n) + (N_t + 2)N_n.$$

MAGI bracket portions (h) For h_{qn} we will use

$$y[q_h(q,n)] = h_{qn}, \tag{4.13}$$

where

$$q_h(q,n) = q_*(C_h, q, n, 0; N_q, N_n - n_m, 1), \tag{4.14}$$

with q_h running from C_h to $C_m - 1$, where

$$C_m = q_C(C_h, N_q, N_n - n_m, 1) = N_i(N_j(N_n + 1) + N_n) + (N_t + 2)N_n + N_q(N_n - n_m).$$

Medicare costs (m) For Medicare costs m_n we will use

$$y[q_m(n)] = m_n, \tag{4.15}$$

where

$$q_m(n) = q_*(C_m, n, 0, 0; N_n, 1, 1) = C_m + n, \tag{4.16}$$

with q_m running from C_m to $C_s - 1$, where

$$C_s = q_C(C_m, N_n, 1, 1) = N_i(N_j(N_n + 1) + N_n) + (N_t + 3)N_n + N_q(N_n - n_m).$$

LTCG bracket allocations (q) For LTCG bracket allocation variables q_{pn} we will use

$$y[q_q(p, n)] = q_{pn}, \tag{4.17}$$

where

$$q_q(p, n) = q_*(C_q, p, n, 0; N_p, N_n, 1), \tag{4.18}$$

with q_q running from C_q to $C_s - 1$, where

$$C_q = q_C(C_m, N_n, 1, 1) = N_i(N_j(N_n + 1) + N_n) + (N_t + 3)N_n + N_q(N_n - n_m),$$

$$C_s = q_C(C_q, N_p, N_n, 1) = N_i(N_j(N_n + 1) + N_n) + (N_t + 3 + N_p)N_n + N_q(N_n - n_m).$$

Surplus (s) Surplus can be generated if big-ticket items are received (inheritance, sale of a house, etc.) or due to RMDs. Surplus s is then deposited to taxable savings accounts according to variable η. We will use

$$y[q_s(n)] = s_n, \tag{4.19}$$

where

$$q_s(n) = q_*(C_s, n, 0, 0; N_n, 1, 1) = C_s + n, \tag{4.20}$$

with q_s running from C_s to $C_w - 1$, where

$$C_w = q_C(C_s, N_n, 1, 1) = N_i(N_j(N_n + 1) + N_n) + (N_t + 4 + N_p)N_n + N_q(N_n - n_m).$$

Withdrawals (w) For withdrawals w_{ijn} we will use

$$y[q_w(i, j, n)] = w_{ijn}, \tag{4.21}$$

where

$$q_w(i, j, n) = q_*(C_w, i, j, n; N_i, N_j, N_n), \tag{4.22}$$

with q_w running from C_w to $C_x - 1$, where

$$C_x = q_C(C_w, N_i, N_j, N_n) = N_i(N_j(2N_n + 1) + (N_t + 4 + N_p)N_n) + N_q(N_n - n_m).$$

Roth conversions (x) Finally, for Roth conversions x_{in} we will use

$$y[q_x(i,n)] = x_{in}, \tag{4.23}$$

where

$$q_x(i,n) = q_*(C_x, i, n, 0; N_i, N_n, 1), \tag{4.24}$$

with q_x running from C_x to $C_* - 1$, where

$$C_* = q_C(C_x, N_i, N_n, 1) = N_i(N_j(2N_n + 1) + (N_t + N_i + 3 + N_p)N_n). \tag{4.25}$$

With $N_i = 2, N_j = 3, N_k = 4, N_t = 7, N_p = 3$ we have $(27 + N_p)N_n + 6 = 30N_n + 6$ variables. The $N_pN_n = 3N_n$ additional variables are the LTCG bracket allocations q_{pn}. When IRMAA is optimized, h_{qn} adds $N_q(N_n - n_m)$ more variables. If the time resolution is increased to months, the number of variables scales accordingly. Adding binary variables z^x_{nz} with $z \in \{0, 1, 2, 3\}$ can add $4N_n$ more decision variables, while adding z^m_{qn} adds $N_q(N_n - n_m)$ more, resulting in $(30N_n + 6) + N_q(N_n - n_m) + 4N_n + N_q(N_n - n_m)$ variables.

4.1 Reverse mapping of indices

The inverse functions for the index-mapping functions will be derived for the most complex case encountered in this paper. If we have

$$z = q_*(C, i, j, k, n; N_i, N_j, N_k, N_n) := C + iN_jN_kN_n + jN_kN_n + kN_n + n, \tag{4.26}$$

then $(i, j, k, n) = q_*^{-1}(z; N_i, N_j, N_k, N_n, C)$ is obtained from

$$\begin{aligned} n &= \mathtt{mod}(\mathtt{mod}(\mathtt{mod}(z - C, N_jN_kN_n), N_kN_n), N_n), \\ k &= \mathtt{mod}(\mathtt{mod}(z - C - n, N_jN_kN_n), N_kN_n)/N_n, \\ j &= \mathtt{mod}(z - C - n - kN_n, N_jN_kN_n)/(N_kN_n), \\ i &= (z - C - n - kN_n - jN_kN_n)/(N_jN_kN_n). \end{aligned} \tag{4.27}$$

While this holds for all cases presented in the previous section, this can be easily simplified for cases having fewer active indices. However, some modern languages can accomplish this mapping rather easily by providing `reshape()` functions.

5. Building constraint matrices

Let's first define generic index-mapping functions I and J as

$$\begin{aligned} I_l(n) &= C_l + n, \\ I_l(i, n; N_n) &= C_l + iN_n + n, \\ I_l(i, j, n; N_j, N_n) &= C_l + iN_jN_n + jN_n + n, \\ \dots &= \dots \end{aligned} \tag{5.1}$$

and so on, which would cumulatively increase row count C_l at each new instance l, similar to how we proceeded in the previous section. This allows us to build rectangular matrices by iteratively adding rows. These constraint matrices have C_* (defined in Eq. (4.25)) columns but can have less rows, forming an underdetermined system to be optimized using linear programming. Function J is defined similarly for equality constraints, while I is used for building the rows of the matrix containing the inequality constraints. Additional indices found in columns and not in rows implies the existence of multiple elements on that row.

5.1 Inequality constraints

Inequality constraints can be upper or lower bounds expressed in matrix form as $\ell \leq A_u y \leq u$. When ℓ is not specified it is assumed 0, and an unspecified u implies ∞. Most if not all inequalities will be expressed as upper bounds.

Required minimum distributions (RMDs) We rewrite the inequality constraint on required minimum distributions Eq. (3.5) using matrix $A_u y \leq u$ starting with the following $N_i N_n$ rows,

$$\begin{aligned} A_u[I_0(i,n), q_w(i,1,n)] &= -1 \\ A_u[I_0(i,n), q_b(i,1,n)] &= \rho_{in}, \\ u[I_0(i,n)] &= 0, \\ & \qquad \forall i \in \{0, \dots, N_i - 1\}, \\ & \qquad \forall n \in \{0, \dots, N_n - 1\}, \end{aligned} \tag{5.2}$$

and all other elements in the same rows of A_u being 0. Notice that while b has $N_n + 1$ elements, the constraints for b go from 0 to $N_n - 1$ as there is no RMD required in the last year of the plan N_n, when all individuals have passed. Years before RMDs where $\rho_{in} = 0$ just add a trivial constraint. See Eq. (4.4).

Ordinary income tax brackets Similarly, we add $N_t N_n$ more rows to matrix $A_u y \leq u$ to express the inequality constraint in Eq. (3.6) setting an upper limit on amounts f_{tn},

$$\begin{aligned} A_u[I_1(t,n), q_f(t,n)] &= 1, \\ u[I_1(t,n)] &= \bar{\Delta}_{tn}, \\ & \qquad \forall t \in \{0,\ldots,N_t-1\}, \\ & \qquad \forall n \in \{0,\ldots,N_n-1\}, \end{aligned} \tag{5.3}$$

and all other elements in the same rows of A_u being 0.

Standard exemption For $e_n \leq \bar{\sigma}_n$ we add

$$\begin{aligned} A_u[I_2(n), q_e(n)] &= 1, \\ u[I_2(n)] &= \bar{\sigma}_n, \\ & \qquad \forall n \in \{0,\ldots,N_n-1\}. \end{aligned} \tag{5.4}$$

Non-negativity of e_n provides the lower bound.

Withdrawal limits For $w_{ijn} + \delta(j,1)x_{in} \leq b_{ijn}$ we add:

$$\begin{aligned} A_u[I_3(i,j,n), q_w(i,j,n)] &= 1, \\ A_u[I_3(i,j,n), q_x(i,n)] &= \delta(j,1), \\ A_u[I_3(i,j,n), q_b(i,j,n)] &= -1, \\ u[I_3(i,j,n)] &= 0, \\ & \qquad \forall i \in \{0,\ldots,N_i-1\}, \\ & \qquad \forall j \in \{0,\ldots,N_j-1\}, \\ & \qquad \forall n \in \{0,\ldots,N_n-1\} \ (n < n_d \text{ for } i = i_d). \end{aligned}$$

For a couple, the upper index for the first-to-pass individual i_d is $n_d - 1$ rather than $N_n - 1$, since posthumous constraints zero out all account activities.

Posthumous account activities For $n \geq n_d$ and the first-to-pass individual i_d, we set

$$\begin{aligned} A_u[I_4(j,n), q_w(i_d,j,n)] &= 1, \\ u[I_4(j,n)] &= 0, \\ A_u[I_5(n), q_d(i_d,n)] &= 1, \\ u[I_5(n)] &= 0, \\ A_u[I_6(n), q_x(i_d,n)] &= 1, \\ u[I_6(n)] &= 0, \\ & \qquad \forall j \in \{0,\ldots,N_j-1\}, \\ & \qquad \forall n \in \{n_d,\ldots,N_n-1\}. \end{aligned}$$

Roth conversion limit If a maximum conversion x_{max} is specified, we add the bound $\sum_i x_{in} \leq x_{max}$ for all n with

$$\begin{aligned} A_u[I_7(n), q_x(i,n)] &= 1, \\ u[I_7(n)] &= x_{max}, \\ & \qquad \forall i \in \{0, \ldots, N_i - 1\}, \\ & \qquad \forall n \in \{0, \ldots, N_n - 1\}. \end{aligned}$$

This maximum is applied to the combined conversions of both individuals in each year.

Exclusion constraints The at-most-one (AMO) exclusions of Eqs. (3.16) and (3.17) are implemented with big-$\mathcal{M}$ constraints using binary variables z^x_{nz}, adding the corresponding rows for $\sum_i w_{i0n} + \sum_i w_{i2n}$, s_n, and $\sum_i x_{in}$ with upper bounds $\mathcal{M} z_{nz}$, along with the bounds $0 \leq z^x_{n0} + z^x_{n1} \leq 1$ and $0 \leq z^x_{n2} + z^x_{n3} \leq 1$.

$$\begin{aligned} A_u[I_9(n), q_w(i,0,n)] &= 1, \\ A_u[I_9(n), q_w(i,2,n)] &= 1, \\ A_u[I_9(n), q_{z^x}(n,0)] &= -\mathcal{M}, \\ u[I_9(n)] &= 0, \\ A_u[I_{10}(n), q_s(n)] &= 1, \\ A_u[I_{10}(n), q_{z^x}(n,1)] &= -\mathcal{M}, \\ u[I_{10}(n)] &= 0, \\ A_u[I_{11}(n), q_{z^x}(n,0)] &= 1, \\ A_u[I_{11}(n), q_{z^x}(n,1)] &= 1, \\ u[I_{11}(n)] &= 1, \\ A_u[I_{12}(n), q_x(i,n)] &= 1, \\ A_u[I_{12}(n), q_{z^x}(n,2)] &= -\mathcal{M}, \\ u[I_{12}(n)] &= 0, \\ A_u[I_{13}(n), q_w(i,2,n)] &= 1, \\ A_u[I_{13}(n), q_{z^x}(n,3)] &= -\mathcal{M}, \\ u[I_{13}(n)] &= 0, \\ A_u[I_{14}(n), q_{z^x}(n,2)] &= 1, \\ A_u[I_{14}(n), q_{z^x}(n,3)] &= 1, \\ u[I_{14}(n)] &= 1, \\ & \qquad \forall i \in \{0, \ldots, N_i - 1\}, \\ & \qquad \forall n \in \{0, \ldots, N_n - 1\}. \end{aligned}$$

Medicare brackets Medicare's MAGI brackets use binary variables z^m_{qn} and the disaggregated MAGI variables h_{qn}. We set upper bounds for each bracket using Eq. (3.42) as

$$\begin{aligned} A_u[I_{15}(q,n), q_h(q,n)] &= 1, \\ A_u[I_{15}(q,n), q_{z^m}(q,n)] &= -\bar{\mathcal{L}}_{qn}, \\ u[I_{15}(q,n)] &= 0, \\ & \qquad \forall q \in \{0,\ldots,N_q-2\}, \\ & \qquad \forall n \in \{n_m,\ldots,N_n-1\}, \end{aligned} \tag{5.5}$$

the upper bound for the last bracket ($q = N_q - 1$) using big-$\mathcal{M}$ as

$$\begin{aligned} A_u[I_{15}(N_q-1,n), q_h(N_q-1,n)] &= 1, \\ A_u[I_{15}(N_q-1,n), q_{z^m}(N_q-1,n)] &= -\mathcal{M}\gamma_n, \\ u[I_{15}(N_q-1,n)] &= 0, \\ & \qquad \forall n \in \{n_m,\ldots,N_n-1\}, \end{aligned} \tag{5.6}$$

and the lower bounds as

$$\begin{aligned} A_u[I_{16}(q,n), q_h(q,n)] &= -1, \\ A_u[I_{16}(q,n), q_{z^m}(q,n)] &= \bar{\mathcal{L}}_{(q-1)n}, \\ u[I_{16}(q,n)] &= 0, \\ & \qquad \forall q \in \{1,\ldots,N_q-1\}, \\ & \qquad \forall n \in \{n_m,\ldots,N_n-1\}. \end{aligned} \tag{5.7}$$

5.2 Equality constraints

Account balances For the equality constraints on account balances expressed in Eq. (3.20), we define an equality constraint matrix $A_e y = v$ starting with $N_i N_j N_n$ rows as

$$\begin{aligned} A_e[J_0(i,j,n), q_b(i,j,n+1)] &= 1, \\ A_e[J_0(i,j,n), q_b(i,j,n)] &= -(1+\mathcal{T}_{ijn}), \\ A_e[J_0(i,j,n), q_x(i,n)] &= -(\delta(j,2)-\delta(j,1))(1+\mathcal{T}_{ijn}), \\ A_e[J_0(i,j,n), q_w(i,j,n)] &= (1+\mathcal{T}_{ijn}), \\ A_e[J_0(i,j,n), q_d(i,n)] &= -\delta(j,0)(1+\mathcal{T}_{ijn}), \\ & \qquad \forall i \in \{0,\ldots,N_i-1\}, \\ & \qquad \forall j \in \{0,\ldots,N_j-1\}, \\ & \qquad \forall n \in \{0,\ldots,N_n-1\}, \end{aligned} \tag{5.8}$$

where v is

$$v[J_0(i,j,n)] = \kappa_{ijn}(1+\mathcal{T}_{ijn}/2). \tag{5.9}$$

The initial account balances expressed in Eq. (3.13) are imposed through

$$\begin{aligned} A_e[J_1(i,j), q_b(i,j,0)] &= 1, \\ v[J_1(i,j)] &= \beta_{ij}, \end{aligned} \tag{5.10}$$

$$\begin{aligned} &\forall i \in \{0,\ldots,N_i-1\}, \\ &\forall j \in \{0,\ldots,N_j-1\}, \end{aligned} \tag{5.11}$$

leading to $N_i N_j$ additional rows to A_e.

Medicare bracket selection The SOS1 selection of Eq. (3.40) is encoded as

$$\begin{aligned} A_e[J_2(n), q_{z^m}(q,n)] &= 1, \\ v[J_2(n)] &= 1, \\ &\forall q \in \{0,\ldots,N_q-1\}, \\ &\forall n \in \{n_m,\ldots,N_n-1\}. \end{aligned} \tag{5.12}$$

MAGI disaggregation The equality $\sum_q h_{qn} = \mathbb{G}_{(n-2)}$ is expanded using Eqs. (2.18) and (2.19) and implemented as

$$\begin{aligned} A_e[J_3(n), q_h(q,n)] &= 1, \\ A_e[J_3(n), q_e(n-2)] &= -1, \\ A_e[J_3(n), q_w(i,1,n-2)] &= -1, \\ A_e[J_3(n), q_x(i,n-2)] &= -1, \\ A_e[J_3(n), q_b(i,0,n-2)] &= -\mu\alpha_{i00(n-2)} - \sum_{k\neq 0} \alpha_{i0k(n-2)}\tau_{k(n-2)}, \\ A_e[J_3(n), q_d(i,n-2)] &= -\mu\alpha_{i00(n-2)} - \sum_{k\neq 0} \alpha_{i0k(n-2)}\tau_{k(n-2)}, \\ A_e[J_3(n), q_w(i,0,n-2)] &= \mu\alpha_{i00(n-2)} + \sum_{k\neq 0} \alpha_{i0k(n-2)}\tau_{k(n-2)} - \alpha_{i00(n-2)} \max(0, \tau_{0\max(0,n-3)} - \mu), \\ v[J_3(n)] &= \mathcal{A}^x_{n-2} + \mathcal{A}^c_{n-2} + \sum_i \left[\omega_{i(n-2)} + \upsilon_{i(n-2)} + \bar{\zeta}_{i(n-2)} + \bar{\pi}_{i(n-2)}\right] \\ &\quad +0.5 \sum_i \left[\kappa_{i0(n-2)} \left(\mu\alpha_{i00(n-2)} + \sum_{k\neq 0} \alpha_{i0k(n-2)}\tau_{k(n-2)}\right)\right], \\ &\quad \forall i \in \{0,\ldots,N_i-1\}, \\ &\quad \forall k \in \{0,\ldots,N_k-1\}, \\ &\quad \forall q \in \{0,\ldots,N_q-1\}, \\ &\quad \forall n \in \{\max(n_m,2),\ldots,N_n-1\}. \end{aligned} \tag{5.13}$$

When $n_m < 2$, we instead set $v[J_3(n)] = \mathbb{G}_{(-1)}$ and $\mathbb{G}_{(-2)}$, which are user-provided MAGI values for the years before the plan started.

Medicare costs Binary variables z^m_{qn} express the active IRMAA bracket. The costs for Medicare are obtained using Eq. (3.43):

$$\begin{aligned} A_e[J_4(n), q_m(n)] &= 1, \\ A_e[J_4(n), q_{z^m}(q,n)] &= -\bar{\mathcal{C}}_{qn}, \\ v[J_4(n)] &= 0, \\ & \qquad \forall q \in \{0, \ldots, N_q - 1\}, \\ & \qquad \forall n \in \{n_m, \ldots, N_n - 1\}. \end{aligned}$$

Net spending For the equality constraint on net spending expressed in Eq. (3.21), we add N_n more rows to $A_e y = v$ as

$$\begin{aligned} A_e[J_5(n), q_g(n)] &= 1, \\ A_e[J_5(n), q_m(n)] &= 1, \\ A_e[J_5(n), q_s(n)] &= 1, \\ A_e[J_5(n), q_f(t,n)] &= \theta^x_{tn}, \\ A_e[J_5(n), q_w(i,j,n)] &= -1 + 0.1\,\delta(j,1)\,(1 - \mathcal{H}(n - n_{i,59\frac{1}{2}})), \\ A_e[J_5(n), q_q(1,n)] &= 0.15, \\ A_e[J_5(n), q_q(2,n)] &= 0.20, \\ & \qquad \forall t \in \{0, \ldots, N_t - 1\}, \\ & \qquad \forall i \in \{0, \ldots, N_i - 1\}, \\ & \qquad \forall j \in \{0, \ldots, N_j - 1\}, \\ & \qquad \forall n \in \{0, \ldots, N_n - 1\}, \end{aligned}$$

where $q_q(p,n)$ is the flat index of q_{pn} in the solution vector, and v is

$$v[J_5(n)] = \sum_{*=x,c,f} \mathcal{A}^*_n + \sum_i \Lambda^{\pm}_{in} + \sum_i [\omega_{in} + \upsilon_{in} + \bar{\zeta}_{in} + \bar{\pi}_{in}] - \mathcal{D}_n. \tag{5.14}$$

The condition of having a predictable net spending expressed as an equality in Eq. (3.26) adds $N_n - 1$ more rows to $A_e y = v$ as

$$\begin{aligned} A_e[J_6(n), q_g(0)] &= -\bar{\xi}_n, \\ A_e[J_6(n), q_g(n)] &= \xi_0, \\ v[J_6(n)] &= 0, \\ & \qquad \forall n \in \{1, \ldots, N_n\}. \end{aligned} \tag{5.15}$$

Taxable ordinary income For the equality constraint in Eq. (3.29) establishing taxable ordinary income, we add N_n rows to $A_e y = v$ as follows. The taxable Social Security term $\Psi_n \sum_i \bar{\zeta}_{in}$ appears

on the right-hand side as a parameter updated each SC-loop iteration.

$$
\begin{aligned}
A_e[J_7(n), q_e(n)] &= 1, \\
A_e[J_7(n), q_f(t,n)] &= 1, \\
A_e[J_7(n), q_w(i,1,n)] &= -1, \\
A_e[J_7(n), q_x(i,n)] &= -1, \\
A_e[J_7(n), q_b(i,0,n)] &= -\sum_{k\neq 0} \alpha_{i0kn}\tau_{kn}, \\
A_e[J_7(n), q_w(i,0,n)] &= \sum_{k\neq 0} \alpha_{i0kn}\tau_{kn}, \\
A_e[J_7(n), q_d(i,n)] &= -\sum_{k\neq 0} \alpha_{i0kn}\tau_{kn}, \\
&\quad \forall t \in \{0,\ldots,N_t-1\}, \\
&\quad \forall i \in \{0,\ldots,N_i-1\}, \\
&\quad \forall k \in \{0,\ldots,N_k-1\}, \\
&\quad \forall n \in \{0,\ldots,N_n-1\},
\end{aligned}
\tag{5.16}
$$

with

$$
v[J_7(n)] = \mathcal{A}_n^x + \Psi_n \sum_i \bar{\zeta}_{in} + \sum_i [\omega_{in} + \upsilon_{in} + \bar{\pi}_{in}] + 0.5 \sum_i \kappa_{i0n} \sum_{k\neq 0} \alpha_{i0kn}\tau_{kn}. \tag{5.17}
$$

5.3 Other considerations

Beneficiaries Tax-free and tax-deferred accounts have special tax rules that allow giving part or the entire value of tax-free accounts to a spouse who can then consider it as his/her own. These accounts typically use percentages to designate beneficiaries. Let ϕ_j be the fraction of the account j that a spouse i_d wishes to leave to his/her surviving spouse i_s in the year $n_d - 1 < N_n - 1$ following the year of passing. To account for that event in year n_d, Eq. (3.20) needs to be rewritten as

$$
\begin{aligned}
b_{ij(n+1)} = {} & [1 - \delta(n, n_d - 1)\delta(i, i_d)] \\
& \times \Big\{ [b_{ijn} + \delta(j,0)d_{ij} - w_{ijn} + (\delta(j,2) - \delta(j,1))x_{in}]\,(1 + \mathcal{T}_{ijn}) + \kappa_{ijn}(1 + \mathcal{T}_{ijn}/2) \Big\} \\
& + [\phi_j \delta(n, n_d - 1)\delta(i, i_s)] \\
& \times \Big\{ [b_{i_djn} + \delta(j,0)d_{i_dn} - w_{i_djn} + (\delta(j,2) - \delta(j,1))x_{i_dn}]\,(1 + \mathcal{T}_{i_djn}) \\
& + \kappa_{i_djn}(1 + \mathcal{T}_{i_djn}/2) \Big\}.
\end{aligned}
\tag{5.18}
$$

The first multiplier [...] on the right-hand side will always be one except for i_d in year $n_d - 1$ when it will be zero. This will result in emptying all accounts for i_d for years n_d and beyond. The second special multiplier [...] before the second set of curly braces {} will always be zero except for the surviving spouse i_s in year $n_d - 1$, who will then inherit a fraction ϕ_j of account j that was scheduled to go into i_d's j account at the beginning of year n_d.

Rewriting the last equation as a constraint results in

$$
\begin{aligned}
& b_{ij(n+1)} \\
& -[1-\delta(n,n_d-1)\delta(i,i_d)] \\
& \times\Big\{[b_{ijn}+\delta(j,0)d_{in}-w_{ijn}+(\delta(j,2)-\delta(j,1))x_{ikn}]\,(1+\mathcal{T}_{ijn})\Big\} \\
& -[\phi_j\delta(n,n_d-1)\delta(i,i_s)] \\
& \times\Big\{[b_{i_djn}+\delta(j,0)d_{in}-w_{ijn}+(\delta(j,2)-\delta(j,1))x_{i_dkn}]\,(1+\mathcal{T}_{i_djn})\Big\} \\
= \; & [1-\delta(n,n_d-1)\delta(i,i_d)]\kappa_{ijn}(1+\mathcal{T}_{ijn}/2) \\
& +[\phi_j\delta(n,n_d-1)\delta(i,i_s)]\kappa_{i_djn}(1+\mathcal{T}_{i_djn}/2). \qquad (5.19)
\end{aligned}
$$

We are now ready to replace Eq. (5.8) for $A_e y = v$ by

$$
\begin{aligned}
A_e[J_0(i,j,n),q_b(i,j,n+1)] &= 1, \\
A_e[J_0(i,j,n),q_b(i,j,n)] &= -[1-\delta(n,n_d-1)\delta(i,i_d)](1+\mathcal{T}_{ijn}), \\
A_e[J_0(i,j,n),q_d(i,j,n)] &= -[1-\delta(n,n_d-1)\delta(i,i_d)]\delta(j,0)(1+\mathcal{T}_{ijn}), \\
A_e[J_0(i,j,n),q_w(i,j,n)] &= [1-\delta(n,n_d-1)\delta(i,i_d)](1+\mathcal{T}_{ijn}), \\
A_e[J_0(i,j,n),q_x(i,n)] &= -[1-\delta(n,n_d-1)\delta(i,i_d)](\delta(j,2)-\delta(j,1))(1+\mathcal{T}_{ijn}), \\
\text{when } N_i=2 \text{ and } i=i_s, & \\
A_e[J_0(i,j,n),q_b(i_d,j,n)] &= -[\phi_j\delta(n,n_d-1)\delta(i,i_s)](1+\mathcal{T}_{i_djn}), \\
A_e[J_0(i,j,n),q_d(i_d,n)] &= -[\phi_j\delta(n,n_d-1)\delta(i,i_s)]\delta(j,0)(1+\mathcal{T}_{i_djn}), \\
A_e[J_0(i,j,n),q_w(i_d,j,n)] &= [\phi_j\delta(n,n_d-1)\delta(i,i_s)](1+\mathcal{T}_{i_djn}), \\
A_e[J_0(i,j,n),q_x(i_d,n)] &= -[\phi_j\delta(n,n_d-1)\delta(i,i_s)](\delta(j,2)-\delta(j,1))(1+\mathcal{T}_{i_djn}),
\end{aligned}
$$

$$
\begin{aligned}
&\forall i\in\{0,\ldots,N_i-1\}, \\
&\forall j\in\{0,\ldots,N_j-1\}, \\
&\forall n\in\{0,\ldots,N_n-1\},
\end{aligned}
$$

where v is

$$
\begin{aligned}
v[J_0(i,j,n)] \;=\; & [1-\delta(n,n_d-1)\delta(i,i_d)]\kappa_{ijn}(1+\mathcal{T}_{ijn}/2) \\
& +[\phi_j\delta(n,n_d-1)\delta(i,i_s)]\kappa_{i_djn}(1+\mathcal{T}_{i_djn}/2). \qquad (5.20)
\end{aligned}
$$

While the last two equations may look cumbersome, their net effect is only to include a few more terms when $n = n_d - 1$.

Assets allocation ratios When asset allocation ratios α are imposed, they should also be applied to how contributions amounts κ_{ijn} are invested, such that

$$\kappa_{ijkn} = \alpha_{ijkn}\kappa_{ijn}. \qquad (5.21)$$

For other allocation schemes, just substitute $\alpha_{ijkn} = \alpha_{ikn}$ or α_{kn} depending on the scheme selected.

Assets allocation have been handled easily by assuming that the accounts are always rebalanced and only using a single multiplier $\mathcal{T}$, defined as

$$\mathcal{T}_{ijn} = \sum_k \alpha_{ijkn} \tau_{kn}, \tag{5.22}$$

to compute to return on the total balance of each savings account.

Spousal deposits and withdrawals While keeping the problem linear, a simple constraint can be imposed on surplus deposits in taxable savings accounts. One can specify a spousal ratio η such as

$$d_{0n} = \eta d_{1n}. \tag{5.23}$$

A similar spousal ratio can be imposed on withdrawals from tax-deferred accounts

$$w_{01n} = \eta w_{11n}, \tag{5.24}$$

but this can cause drawing from an empty account while the other spousal account is not. Only the deposit scheme has currently been implemented in Owl.

6. Objective functions

The objective function is a simple scalar defined as $c \cdot y$ that will be minimized.

Maximum net spending There are a few ways by which a retirement plan can be optimized. For maximizing the net spending under the constraint of a desired bequest, we introduce the following relation

$$E_n = \sum_{i,j} (1 - \nu\delta(j,1)) b_{ijn}, \tag{6.1}$$

which is the value of the estate in nominal dollars at year n, taking into consideration the heir's marginal income tax rate ν on the $(j = 1)$ tax-deferred account.

For a desired bequest ϵ_{N_n}, expressed in today's dollars, the final amount in year N_n will need to be

$$E_{N_n} = \bar{\epsilon}_{N_n} = \epsilon_{N_n} \gamma_{N_n}. \tag{6.2}$$

Fixing a bequest value amounts to adding the following constraint

$$\sum_{i,j} b_{ijN_n} (1 - \nu\delta(j,1)) = \epsilon_{N_n} \gamma_{N_n}, \tag{6.3}$$

which would add one more row to $A_e y = v$ as

$$\begin{aligned} A_e[I(0), q_b(i,j,N_n)] &= (1 - \nu\delta(j,1)) \\ v[I(0)] &= \epsilon_{N_n} \gamma_{N_n} \qquad (6.4) \\ &\qquad \forall i \in \{0, \ldots, N_i - 1\}, \\ &\qquad \forall j \in \{0, \ldots, N_j - 1\}, \qquad (6.5) \end{aligned}$$

where $I(0)$ is used only to provide the proper row offset C_l. See Eq. (5.1).

For maximizing the net spending under the constraint of a fixed bequest, one may either maximize the first-year net spending basis g_0 (with the profile fixed by Eq. (3.26)), or maximize the sum of net spending over all years in today's dollars. In the first case, minimize the inner product $c \cdot y$ with

$$c[q_g(0)] = -1, \tag{6.6}$$

and 0 otherwise; the whole time series g_n then follows from Eq. (3.26). In the second case, when the slack variable λ is used over net spending (as in Eq. (6.10) below), the objective is the sum

$\sum_{n=0}^{N_n-1} g_n/\gamma_n$, maximized by setting

$$c[q_g(n)] \quad = \quad -1/\gamma_n, \tag{6.7}$$

$\forall n \in \{0, \ldots, N_n-1\}$, and 0 otherwise. This sum-of-net-spending formulation is the one implemented in Owl.

Maximum variable net spending When the slack variable λ is used (Eq. (3.27) or Eq. (6.10) below), the objective function is the sum of net spending over the full duration of the plan in today's dollars, $\sum_{n=0}^{N_n-1} g_n/\gamma_n$. The coefficient vector is

$$c[q_g(n)] \quad = \quad -1/\gamma_n, \tag{6.8}$$

$\forall n \in \{0, \ldots, N_n - 1\}$ and 0 otherwise. In that case, constraint equality Eq. (3.26) is replaced by an inequality. Instead of obeying

$$g_n \xi_0 - g_0 \bar{\xi}_n = 0, \tag{6.9}$$

we impose the inequality constraint

$$(1-\lambda) g_0 \bar{\xi}_n / \xi_0 \leq g_n \leq (1+\lambda) g_0 \bar{\xi}_n / \xi_0, \tag{6.10}$$

where λ is the percentage expressed in decimal that the annual net spending is allowed to deviate from the desired profile. It should be noticed that when $\lambda = 0$ the two last equations are equivalent.

Maximum bequest If, on the other hand, one would like to maximize the bequest under the constraint of a desired net spending g_o, specified for the first year, one would add the following row to $A_e y = v$

$$\begin{aligned} A_e[I(0), q_g(0)] &= 1, \\ v[I(0)] &= g_o, \end{aligned} \tag{6.11}$$

subject to the net spending g_n obeying Eq. (3.26) over time.

The objective function would then be derived from Eq. (6.1) as minimizing the inner product $c \cdot y$, where c is

$$c[q_b(i, j, N_n)] \quad = \quad -(1 - \nu\delta(j, 1)), \tag{6.12}$$

$$\forall i \in \{0, \ldots, N_i - 1\}, \quad \forall j \in \{0, \ldots, N_j - 1\}, \tag{6.13}$$

and 0 otherwise.

Lexicographic weight for Roth conversions and preferential withdrawals The coefficient vector c can be augmented with a small positive weight ε to break ties without changing the primary

objective in practice. In addition to the coefficients of the chosen primary objective (max spending or max bequest), the implementation adds the following contributions to c:

$$\begin{aligned} c[q_x(i,n)] &\leftarrow c[q_x(i,n)] + \varepsilon, \\ &\qquad \forall i \in \{0,\ldots,N_i-1\},\ \forall n \in \{0,\ldots,N_n-1\}, \qquad (6.14) \\ c[q_w(1,j,n)] &\leftarrow c[q_w(1,j,n)] + \varepsilon, \\ &\qquad \forall j \in \{0,\ldots,N_j-1\},\ \forall n \in \{0,\ldots,N_n-1\}, \quad \text{when } N_i = 2. \qquad (6.15) \end{aligned}$$

The first line penalizes Roth conversions x_{in}, so that among solutions with the same primary objective value the solver prefers fewer or smaller conversions (reducing churn). The second line penalizes withdrawals w_{1jn} from the second individual ($i = 1$), so that withdrawals from the first individual ($i = 0$) are preferred when otherwise equivalent. The weight ε is chosen small (e.g. of order 10^{-9} in the same units as y) so that the primary objective dominates; it is often enabled by default when optimizing Medicare (IRMAA) to stabilize bracket selection.